








Modelling Financial Markets

Chapter 19: Review and Project Presentations



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When Does an Empirical Result Deserve Trust?



Albert J. Menkveld. **Photo: Me Judice (2015); CC BY 3.0; Wikimedia Commons**

"In science, evidence is generated to test hypotheses in an evidence-generating process (EGP). We claim that EGP variation across researchers adds uncertainty—nonstandard errors (NSEs)."

— *Menkveld et al. (2024)*

- Menkveld et al. (2024): 164 research teams test the same six hypotheses on the same data
  - for market efficiency, the interquartile range (IQR) of the estimates across teams is 6.7 percentage points (pp)
  - the median standard error (SE) implies a Gaussian IQR of only 3.4%, half the 6.7 pp
- On course data, a mini-multiverse of 27 specifications per index shows the same phenomenon
  - IQR of the estimated annual trend: 64.6 pp for the Euro Stoxx 50 and 82.6 pp for the BET, the Bucharest Stock Exchange index
  - benchmark due to sampling alone: 41.3 pp and 43.4 pp, respectively
- The choice of risk model also changes the conclusion: VaR 1% (Value-at-Risk) for the BET over 5 432 evaluation days
  - the Normal model is breached on 2.25% of days, more than twice as often as it should
  - only FHS (filtered historical simulation), at 1.23%, lies in the 95% binomial range [0.74%, 1.27%]
- The chapter reviews Chapters 0–18 through six pitfalls and prepares the team project
  - replication, pre-registration and a specification grid

▸ Central chart: A Mini-Multiverse on Course Data: 27 Paths per Index

What Turns a Result into a Finding? (1/2)

The Chapter in Brief

- The question: when does an empirical result about markets deserve trust?
  - and when does it only reflect the data, the design or luck?
- Part I: the course in one picture, a review of Chapters 0–18
  - six pitfalls
  - one integrated case study, from raw prices of the BET to a backtested value-at-risk (VaR) at 1%
  - BET: the benchmark index of the Bucharest Stock Exchange, BVB
- The main thread: GARCH forecasts volatility, DCC the correlations, VaR and ES turn them into a loss
  - GARCH: generalised autoregressive conditional heteroskedasticity
  - DCC: dynamic conditional correlation
  - ES: expected shortfall, the mean loss beyond VaR
- Part II: the team project and artificial intelligence (AI) in the project
  - a research case study on nonstandard errors
  - the AI discovery loop
  - the project step by step: ▸ Appendix

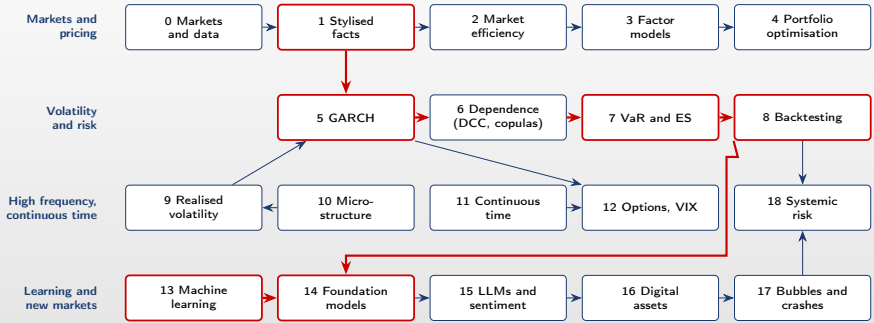
What Turns a Result into a Finding? (2/2)

What you should be able to do

- Compute a backtested risk forecast from the data, with valid inference
- Interpret an empirical result after checking the six pitfalls in your own code
- Compute the power of the main test within a project plan with replication, identification and a specification grid
- Further Topics and Appendix: reading for familiarity

- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

Structure of the Course



What Connects the Steps of the Analysis

- Red: the main thread, from stylised facts to risk forecasts we can test
  - Chapter 1: heavy tails, volatility clustering, almost no linear predictability
  - Chapters 5–8: GARCH gives σ_{t+1}, VaR and ES the loss
  - backtesting says whether to trust it
- Chapters 9–12 measure volatility better and model it in continuous time
  - intraday data, option prices, the VIX
  - VIX: the implied-volatility index of the S&P 500
- Chapters 13–15 bring flexible models, including large language models (LLMs)
  - judged with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system
- Special Chapter 20: realised volatility with path signatures
  - on the VOLARE data (VOLatility Archive for Realized Estimates)
- Further inference formulas (Shanken, QMLE, the empirical quantile): ▸ Appendix

Five Questions for Every Model

Before estimating

- **Data**: which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model**
  - which stylised fact does it capture?
  - which stylised fact does it ignore?
- **Estimation**: on which window, with which information at time t?

After estimating

- **Evaluation**: out of sample, with a loss that matches the target?
  - QLIKE (the quasi-likelihood loss) for variance
  - FZ0 (the Fissler–Ziegel score) for a (VaR, ES) pair
- **Uncertainty**: robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

Chapter 0: Financial Markets in 2026

- Markets combine exchanges, over-the-counter dealers, funds, algorithms and blockchains
  - exchange-traded funds (ETFs)
- SPY, 2015–2026: CAGR 13.7%, annualised volatility 17.6%
  - SPY: S&P 500 ETF, dividends reinvested
  - CAGR: compound annual growth rate
- Bitcoin, same period: CAGR 60.6%, annualised volatility 66.7%
- BET, July 2007 to February 2009: a drawdown of -82.5%
- Standard errors of the mean and the volatility from i.i.d. log returns
  - Y: sample length in years; T: number of daily returns; K: population kurtosis

$$\mathrm{SE}(\hat{\mu}_{\mathbf{ann}}) = \frac{\sigma_{\mathbf{ann}}}{\sqrt{Y}}, \qquad \mathrm{SE}(\hat{\sigma}) \approx \sigma \sqrt{\frac{K-1}{4T}}$$

  - $\mathrm{SE}(\hat{\sigma})$ requires a finite fourth moment, which Chapter 1 questions
- SPY: $\mathrm{SE}(\hat{\mu}_{\mathbf{ann}}) \approx 5.1\%$ for an annualised mean log return of 12.8%
  - means are barely estimable, volatilities are

Chapter 1: Stylised Facts and Returns

- Returns are uncorrelated but not independent Cont (2001)
  - their magnitude is predictable, their sign largely is not
- Tail index α: the probability of a loss above x decays like $x^{-\alpha}$
- The Hill estimator uses the k largest losses, ordered $L_{(1)} \geq \cdots \geq L_{(k+1)}$ Hill (1975)

$$\hat{\alpha} = \Big[\frac{1}{k}\sum_{i=1}^{k} \ln \frac{L_{(i)}}{L_{(k+1)}}\Big]^{-1}, \qquad \mathrm{SE}(\hat{\alpha}) \approx \frac{\hat{\alpha}}{\sqrt{k}} \text{ for i.i.d. losses}$$

- BET, $k = 153$: $\hat{\alpha} = 2.39$, block-bootstrap 95% CI $[2.12, 3.03]$
  - CI: confidence interval
- The i.i.d. interval $\hat{\alpha} \pm 1.96\,\hat{\alpha}/\sqrt{k} = [2.01, 2.77]$ is narrower
  - large losses come in clusters
- $\alpha < 4$: the fourth moment is likely infinite
- Days beyond 5σ are more than 1000 times more frequent than under the Normal distribution
  - the Normal distribution gives $2\Phi(-5) \approx 5.7 \times 10^{-7}$ a day
  - that is, one such day in about 7 000 years

Chapter 2: Market Efficiency

- The efficient market hypothesis: prices fully reflect the information set
- A test of abnormal-return predictability is a joint test Fama (1970)
  - of efficiency and of the expected-return model
- The q-period variance ratio and its heteroskedasticity-robust statistic Lo & MacKinlay (1988)

$$VR(q) = \frac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}, \qquad z^*(q) = \frac{\widehat{VR}(q) - 1}{\widehat{\mathrm{SE}}^*} \Rightarrow N(0,1)$$

  - $z^*(q)$ requires a finite fourth moment, which Chapter 1 questions
- Several horizons at once: $\max_q |z^*(q)|$ with the Chow–Denning critical values Chow & Denning (1993)
- Share of rolling windows that reject the random walk
  - S&P 500 6.7%, BET 22.4%, Bitcoin 0.0%
- Statistical rejection is not exploitable profit

Chapter 3: Factor Models and Asset Pricing

- Expected returns should reward only exposure to systematic factors
- The GRS test that all intercepts (pricing errors) $\hat{\alpha}$ are zero Gibbons, Ross & Shanken (1989)
  - T periods, N assets and K factors

$$\mathrm{GRS} = \frac{T-N-K}{N}\,\frac{\hat{\alpha}^\top \hat{\Sigma}^{-1}\hat{\alpha}}{1+\hat{\mu}_f^\top \hat{\Omega}^{-1}\hat{\mu}_f} \;\sim\; F(N, T-N-K)$$

  - $\hat{\mu}_f$: factor means; $\hat{\Sigma}$, $\hat{\Omega}$: residual and factor covariances
  - the F is exact only under i.i.d. Normal errors
- 25 size and book-to-market portfolios
  - SML (security market line) slope -4.4%
- GRS $= 4.18$, against a 5% critical value below 1.6
  - the CAPM (capital asset pricing model) is rejected ($p < 0.01$)
- Sector ETFs: the first principal component (the market) explains 66% of the variance

Chapter 4: Portfolio Optimisation

- Markowitz asks the right question, but estimation error dominates the answer Markowitz (1952)
- Standard error of the per-period Sharpe ratio, i.i.d. Normal returns over T periods Lo (2002)

$$\mathrm{SE}(\widehat{SR}) \approx \sqrt{(1 + \widehat{SR}^2/2)/T}$$

  - $\mathrm{SE}(\widehat{SR})$ requires a finite fourth moment, which Chapter 1 questions
- 60-month samples: median in-sample Sharpe 1.56
  - median true Sharpe only 0.30
- 96% of the estimated tangency portfolios are truly worse than 1/N DeMiguel, Garlappi & Uppal (2009)
  - 1/N is a hard benchmark
- Portfolio A against 1/N: test $SR_A - SR_B = 0$ with HAC or studentised bootstrap inference Ledoit & Wolf (2008)
  - not by comparing two point estimates

Chapter 5: Conditional Volatility: GARCH

- Volatility clusters, so today's shock predicts tomorrow's variance Engle (1982), Bollerslev (1986)
- The GARCH(1,1) model and its h-day forecast
  - ε_t: the shock; α, β: the weights of the last shock and of the last variance

$$\sigma^2_{t+1} = \omega + \alpha\,\varepsilon^2_t + \beta\,\sigma^2_t, \qquad \bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$$

$$\sigma^2_{T+h|T} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}\left(\sigma^2_{T+1|T} - \bar{\sigma}^2\right)$$

- The persistence $\alpha+\beta$ is the rate at which the forecast returns to the long-run level $\bar{\sigma}^2$
  - the half-life is $h_{1/2} = \ln 0.5 / \ln(\alpha+\beta)$
- S&P 500, 2000–2026: $\alpha+\beta = 0.9948$, so $h_{1/2} = 132$ days
- The BET persistence cannot be bounded away from 1
  - pitfall "Persistence Near 1"

Chapter 6: Multivariate Volatility and Dependence

- Correlations move with the macro regime
  - diversification is not guaranteed
- Crisis correlation rises mechanically with volatility, so test on the adjusted correlation Forbes & Rigobon (2002)

$$\rho^* = \frac{\rho}{\sqrt{1 + \delta(1 - \rho^2)}}$$

  - ρ: crisis correlation
  - δ: relative rise in the variance of the source market from calm to crisis
  - the test on ρ^* requires a finite fourth moment, which Chapter 1 questions
- SPY–TLT: ρ rose from -0.40 to 0.11 after 2022
  - TLT: long US Treasury bond ETF
- Over the same time a 60/40 portfolio went from 10.4% to 13.2% volatility
- Bitcoin–SPY DCC correlation: 0.04 before 2020, 0.31 since

Chapter 7: Value-at-Risk and Expected Shortfall

- VaR is a quantile of the loss
- ES, the average loss beyond it, is coherent
- Definitions, with q_α the α-quantile of the returns r (continuous distribution)

$$\mathrm{VaR}_\alpha = -q_\alpha, \qquad \mathrm{ES}_\alpha = -\mathbb{E}\big[r \mid r \le q_\alpha\big] = \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_u \, du$$

- S&P 500, daily: historical $\mathrm{VaR}_{1\%} = 3.45\%$, $\mathrm{ES}_{1\%} = 5.03\%$
- The Normal distribution gives only 2.80%
  - heavy tails matter
- Filtered historical simulation and conditional EVT react to volatility within days McNeil & Frey (2000)

Chapter 8: Backtesting and Evaluating Risk Forecasts

- Test the number and the timing of breaches: coverage and independence Kupiec (1995), Christoffersen (1998)
- Kupiec, with x breaches in T days and the observed rate $\hat{\pi} = x/T$

$$LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x}\alpha^{x}}{(1-\hat{\pi})^{T-x}\hat{\pi}^{x}} \sim \chi^2_1$$

- Christoffersen: $LR_{ind} \sim \chi^2_1$ compares the breach rate after a quiet day with the rate after a breach
  - formula: Step 4 of the case study
- S&P 500, VaR 1%, 2007–2026: breach rate (target 1%)
  - Normal 2.70%
  - HS (historical simulation) 1.67%
  - FHS (filtered historical simulation) 1.23%
  - GARCH-EVT (GARCH filter with a tail from extreme value theory) 1.15%
- On the S&P 500 and the BET, a GARCH filter plus a flexible tail wins
  - FHS, GARCH-EVT

Chapter 9: Realised Volatility

- Intraday prices measure daily variance (RV, realised variance)
- The log-HAR (heterogeneous autoregressive) model approximates its observed persistence over finite horizons Corsi (2009)

$$\ln RV_{t+\mathbf{1}} = b_{\mathbf{0}} + b_d \ln RV_t + b_w \overline{\ln RV}_t^{(\mathbf{5})} + b_m \overline{\ln RV}_t^{(\mathbf{22})} + u_{t+\mathbf{1}}$$

  - $\overline{\ln RV}_t^{(\mathbf{5})}$, $\overline{\ln RV}_t^{(\mathbf{22})}$: the mean of $\ln RV$ over the last 5 and 22 days
  - a stationary HAR is a finite AR: short memory asymptotically
- SPY: kurtosis 15.73 for open-to-close returns
  - 2.66 after dividing by $\sqrt{RV_t}$
- log-HAR beats GARCH(1,1)-t in QLIKE
  - by 10% for SPY ($p = 0.021$)
  - by 22% for Bitcoin
  - the DM test on QLIKE requires finite moments and weak dependence of the loss differential d_t, not of the returns

Chapter 10: Market Microstructure

- Prices form in limit order books
- The spread pays for processing, inventory and adverse selection
- The Roll spread: the bid–ask bounce
  - makes consecutive changes of the transaction price, Δp_t, negatively correlated

$$s_{\mathbf{Roll}} = 2\sqrt{-\mathrm{Cov}(\Delta p_t, \Delta p_{t-\mathbf{1}})}$$

- When the sample covariance is positive the estimator is undefined
  - report how often
- Amihud ratio (basis points, bp, per 1 million US dollars, USD)
  - SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

Chapter 11: Continuous-Time Models

- Brownian motion is the limit of random walks with ever smaller steps
- Itô's lemma solves the stochastic differential equations of prices
- Normality of the log returns of geometric Brownian motion (GBM)
  - the Jarque–Bera test, with S the sample skewness and K the sample kurtosis

$$JB = \frac{T}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2_2$$

- S&P 500 against GBM: 62 days beyond ± 4 standard deviations, against 0.59 expected
- The Hill index of S&P 500 losses is below 4 (Chapters 1 and 16)
  - the fourth moment is likely infinite
- Jumps: $(RV - BV)$ standardised $\Rightarrow N(0,1)$ Barndorff-Nielsen & Shephard (2006)
  - BV: the bipower variation, robust to jumps

Chapter 12: Options and the Volatility Surface

- Black–Scholes prices options by hedging Black & Scholes (1973)
  - implied volatility: the value of σ extracted from option prices, a risk-neutral quantity, not a forecast of physical volatility
- The (ex-ante) variance risk premium: $E_t^Q[RV_{t,t+\mathbf{21}}] - E_t^P[RV_{t,t+\mathbf{21}}]$
  - $VIX_t^{\mathbf{2}}$ estimates the first term, the risk-neutral expected variance over the next month; the second needs a forecast
  - $RV_{t,t+\mathbf{21}}$: realised variance over the next 21 trading days
- The ex-post realised gap, known only at $t + 21$:

$$VIX_t^{\mathbf{2}} - RV_{t,t+\mathbf{21}}$$

- Consecutive values overlap, so the mean needs Newey–West standard errors with at least 21 lags
  - the standard error of the mean requires a finite fourth moment, which Chapter 1 questions
- 1990–2026: mean VIX 19.4
  - mean realised volatility over the next month 15.3
- The ex-post realised gap is positive on 86% of days, but not in crashes
  - its sign does not identify the sign of the ex-ante premium

Chapter 13: Machine Learning in Finance

- Flexible models, but little signal and much noise López de Prado (2018)
  - the finance-specific part is the validation process
- Shuffled K-fold reports 69.4% accuracy on pure noise
  - leakage: test information leaking into training
- Remedy: purging and an embargo after the test fold
  - purging: dropping training labels that overlap the test fold
- Selection among N strategies: the deflated Sharpe ratio tests $SR > \mathbb{E}[\max_N \widehat{SR}]$ Bailey & López de Prado (2014)
  - the threshold: the expected best estimated Sharpe ratio of N no-skill strategies
  - the threshold grows with N
- S&P 500: a unit-root test with one lag rejects from $d^* = 0.25$
  - an ADF rejection does not prove stationarity: the finite filter has weights summing to a nonzero value (0.178 at $d = 0.25$), so the series keeps a component proportional to the log price
  - its correlation with the log price, 0.98, only illustrates this component
- No model beats the naive rule "always up"

Chapter 14: Deep Learning and Time-Series Foundation Models

- LSTMs (long short-term memory networks) and Transformers learn from sequences
- Foundation models forecast zero-shot
  - without training on the series at hand
- The Diebold–Mariano test on loss differences d_t between two models Diebold & Mariano (1995)
  - $\hat{\Omega}$: the HAC long-run variance of d_t

$$t_{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}} \Rightarrow N(0,1)$$

- SPY realised volatility, QLIKE (lower is better)
  - Chronos-2 0.223, log-HAR 0.226, GARCH-t 0.273, on the exact grid of 78 five-minute bars
- Chronos-2 and log-HAR are not significantly different (DM $p = 0.539$)
  - the DM test on QLIKE requires finite moments and weak dependence of the loss differential d_t, not of the returns
- Chronos-2 VaR 1% without recalibration on the S&P 500: 1.67% breaches
- With many candidate models: the model confidence set (MCS) Hansen, Lunde & Nason (2011)

Chapter 15: LLMs and Sentiment Analysis

- From dictionaries to FinBERT and LLMs: text becomes a number
- Predictive regression $r_{t+\mathbf{1}} = a + b\,S_t + \varepsilon_{t+\mathbf{1}}$, S_t the sentiment score Stambaugh (1999)
  - a persistent S_t whose innovations correlate with returns biases $\hat{b}$ and its t-test
- Classifiers scored on the same texts are compared as pairs, with the exact McNemar test
- Qwen 7B vs 1.5B on financial headlines: $+21.5$ percentage points in accuracy
  - McNemar $p < 0.001$
- A headline-sentiment strategy: Sharpe $+0.63$ before costs
  - and -1.48 after costs
- Break-even cost: 1.5 bp per trade

Chapter 16: Digital Assets and DeFi

- The chapter case study (Pele et al. (2023)): cryptocurrencies form a distinct asset class
  - in a feature space where a tail factor separates them
- DeFi (decentralised finance) runs on smart contracts
- Bitcoin–S&P 500 link: sup-Wald 20.1 Andrews (1993)
  - sup-Wald: the largest Wald statistic over candidate break dates
  - 5% critical value 11.6
  - the Wald statistic requires $\mathbb{E}[x_t^2 u_t^2] < \infty$ (regressor $\times$ error), a fourth-order condition that Chapter 1 questions
- Estimated break: 9 Mar 2020, not the 2024 ETF launch
- Hill tail index of losses
  - Bitcoin 2.73, 95% CI $[2.12, 3.33]$
  - S&P 500 2.87, 95% CI $[2.07, 3.67]$
- The Bitcoin $-$ S&P 500 difference, from a joint block bootstrap: -0.14, 95% CI $[-1.68, 0.71]$
  - the interval contains 0: no evidence of heavier Bitcoin tails; a non-rejection is not equality
  - the overlap of the two intervals is not a test of the difference

Chapter 17: Bubbles and Crashes

- A bubble is a price path that grows faster than fundamentals can justify
- GSADF: the largest right-tailed ADF (augmented Dickey–Fuller) statistic Phillips, Shi & Yu (2015)
  - over all sample fractions $[r_1, r_2]$ of length at least r_0

$$GSADF = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2}$$

- Volatility shifts distort the size
  - use wild-bootstrap critical values
- Bitcoin 2014–2026 (weekly): GSADF 3.85 against a 95% wild-bootstrap critical value of 2.82 (2.27 Monte Carlo)
  - explosive episodes are detected
- One vs two Markov regimes for Bitcoin: $LR = 123.2$
  - against a parametric-bootstrap 95% critical value of 11.8 ($p = 0.005$)

Chapter 18: Systemic Risk and Networks

- Systemic risk: losses of one institution that spread to the whole system
- ΔCoVaR: the change in the system's VaR when institution i moves from its median to distress
  - X^i: the return of institution i

$$\Delta\text{CoVaR}^i = \text{CoVaR}_\alpha^{sys|X^i=-\text{VaR}_\alpha^i} - \text{CoVaR}_\alpha^{sys|X^i=\textbf{median}}$$

- ΔCoVaR 1% of eleven banks: 4 of 25 within-region pairs differ at 5%
  - 0 after the Holm correction
- Overlap of intervals alone would have ranked none
- LRMES, simulated from GJR–GARCH and DCC Brownlees & Engle (2017)
  - LRMES: the expected loss of a bank when the market falls by more than 10% over 22 days
  - JPMorgan 10.6% today, 23.7% with the information of 31 March 2020

Pitfalls of Design: Look-Ahead Bias and Overfitting

- **Look-ahead bias**: a full-sample estimate leaks the future
  - any parameter, threshold or scaling estimated on the full sample
  - Examples: full-sample GARCH parameters or standardisation, labels that overlap the test set (Chapter 13; the effect on the BET: case study)
  - Remedy: rolling or expanding windows, purging and embargo, the information set $\mathcal{F}_{t-1}$ written next to every forecast
- **Overfitting**: the in-sample fit rewards noise (Chapter 4)
  - the noise will not repeat out of sample
  - Backtest overfitting: trying many strategies and keeping the best Bailey et al. (2017)
  - The deflated Sharpe ratio corrects for the number of trials and for non-normal returns Bailey & López de Prado (2014)
  - Remedy: the design fixed before the test period, every model tried reported, a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall of Design: Multiple Testing (1/2)

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$
  - the best reaches $|t| \approx 3.06$
  - hence the literature asks for $|t| > 3$ Harvey, Liu & Zhu (2016)
- Calendar anomalies, trading rules, specifications
  - each extra test adds a chance of a false discovery
- Remedy: count your tests and correct

Pitfall of Design: Multiple Testing (2/2)

- With m tests, ordered p-values $p_{(1)} \leq \cdots \leq p_{(m)}$ and level a, reject $H_{(i)}$ if

Bonferroni:	$p_{(i)} \leq a/m$
Holm:	$p_{(j)} \leq a/(m-j+1)$ for every $j \leq i$
Benjamini–Hochberg:	$i \leq \max\{k : p_{(k)} \leq k\, a/m\}$

  - Bonferroni and Holm Holm (1979) control the chance of any false discovery (FWER)
  - Holm rejects at least everything Bonferroni rejects
  - Benjamini–Hochberg (BH) Benjamini & Hochberg (1995) controls the expected share of false discoveries among the rejections (FDR)
  - For forecasts: the model confidence set (MCS) instead of many pairwise tests Hansen, Lunde & Nason (2011)

Pitfall of Data: Aligning Returns Instead of Prices

- Bitcoin trades every day, SPY only on weekdays
  - Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
  - Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- Bitcoin, 2020–2026, on the common days: cumulative return 1 025% (right) against 680% (wrong)
  - Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong alignment hides weekend risk
- Remedy: align prices on common days first
  - one series alone stays on its own calendar

Pitfall of Inference: Heavy Tails Ignored

- Classical standard errors assume independence and finite moments of the quantity analysed
  - variances, autocorrelations and Sharpe ratios: the fourth moment of returns; DM: the moments of the loss differential; Gaussian QMLE: the fourth moment of the standardised innovations
  - BET: $\hat{\alpha} = 2.39$, 95% CI $[2.12, 3.03]$ (Chapter 1): under a Pareto-type tail, consistent with an infinite kurtosis, not a proof
  - The sample excess kurtosis (11.1) estimates no finite parameter if $\alpha < 4$, and a bootstrap interval for it is not valid Athreya (1987): report the tail index instead
- Remedy: HAC errors, block bootstrap, intervals that respect the parameter boundary
  - HAC: heteroskedasticity and autocorrelation consistent
  - block bootstrap where it is valid
- The Newey–West HAC standard error of a mean $\bar{d}$ Newey & West (1987): $\mathrm{SE}(\bar{d}) = \sqrt{\hat{\Omega}/T}$, with

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{L}\Big(1 - \frac{j}{L+1}\Big)\hat{\gamma}_j$$

  - $\hat{\gamma}_j$: the sample autocovariance at lag j; L: the number of lags; the Bartlett weights guarantee $\hat{\Omega} \geq 0$
  - The same long-run variance enters the Diebold–Mariano test (Step 5)

Persistence Near 1: Which Boundary? (1/2)

- $p = \alpha + \beta \leq 1$ is a restriction we choose, not a property of GARCH
  - its interior gives a finite unconditional variance and a finite half-life
  - equality is the IGARCH (integrated GARCH) boundary
- Without it, p slightly above 1 is admissible Nelson (1990)
  - the process can still be strictly stationary, when $E \ln(\alpha z_t^2 + \beta) < 0$
  - but it has no finite variance and no half-life
- BET: $\hat{p} = 0.9907$ is inside the region
  - the difficulty is the **true** p, which may equal 1
- At $p = 1$ the estimator is not approximately Normal Andrews (1999)
  - so the Wald interval [0.974, 1.007] is uninformative under our restriction, rather than "wrong"

Persistence Near 1: Which Boundary? (2/2)

- The likelihood-ratio test of H_0: $p = 1$ has the boundary distribution $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$
  - the 5% critical value is 2.71 instead of 3.84
  - BET: $LR = 1.01$, p-value 0.16, so IGARCH is not rejected
- Profile-likelihood CI within $p \leq 1$: $[0.975, 1]$
  - so the half-life is at least 27 days, with no finite upper bound
- Boundary critical values apply only when the true parameter can sit on the edge of the space you imposed
  - not whenever an estimate is close to some bound

Pitfall of Evaluation: a Loss That Cannot Rank the Target

- A variance forecast f_t is compared with a noisy proxy $\hat{\sigma}_t^2$ (r_t^2 or RV_t)
  - the MSE (mean squared error) loss: $(\hat{\sigma}_t^2 - f_t)^2$
  - QLIKE: $\hat{\sigma}_t^2/f_t - \ln(\hat{\sigma}_t^2/f_t) - 1$
- With a conditionally unbiased proxy, MSE and QLIKE rank forecasts as the true variance would Patton (2011)
- QLIKE depends only on the ratio proxy/forecast, so volatile days weigh less than under MSE
- Chapter 9: log-HAR beats GARCH in QLIKE
  - the MSE comparison is inconclusive
- ES alone has no strictly consistent scoring function on the usual broad classes of distributions
- The pair (VaR, ES) has one Fissler & Ziegel (2016): the FZ0 score (Step 5)
  - with Diebold–Mariano tests Diebold & Mariano (1995)
  - their precision is limited in finite samples Pele & Mazurencu-Marinescu-Pele (2026)
- Remedy: choose the loss before the comparison, and state it in the project report

Integrated Case Study: Methods from Four Chapters (1/2)

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026
  - 6 670 daily log returns
- Log return in %: $r_t = 100 \ln(P_t / P_{t-1})$, loss $L_t = -r_t$
- Removed: duplicate records and weekend-dated rows
  - days with an unchanged close kept

Integrated Case Study: Methods from Four Chapters (2/2)

The five steps

- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student-t innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones
- Formal comparison of the four models
  - FZ0 loss, Diebold–Mariano (DM) tests, model confidence set

Uses of the Symbol α in the Case Study

Where	Meaning of α	Value	Enters
Step 1, Hill	tail index of losses	2.39	$P(L > x)$ decays like $x^{-\alpha}$
Step 2, GARCH	weight of yesterday's squared shock	0.192	$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
Steps 3–4, VaR and Kupiec	VaR level: promised breach probability	1%	LR_{uc}, DQ
Step 5, FZ0 and ES	ES level used to score the pair (VaR, ES)	2.5%	$S(v, e; y)$

- Before substituting a number, ask which α the formula means
  - the Kupiec test needs $\alpha = 0.01$, never 2.39 or 0.192
- The letter p also has two meanings here
  - the GARCH persistence $p = \alpha + \beta$ and a p-value

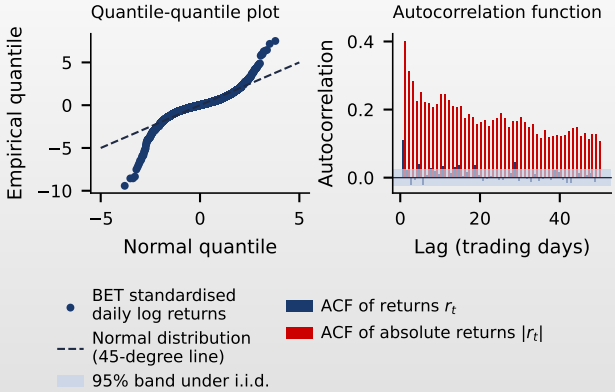
Step 1: What We Measure and How We Test It

- Tails: the Hill tail index $\hat{\alpha}$ of losses, with a block-bootstrap 95% CI
  - formula: the Chapter 1 review
- Volatility clustering: the autocorrelations of $|r_t|$ and the Ljung–Box test on r_t^2

$$Q(m) = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k} \;\sim\; \chi^2_m \text{ under } H_0$$

  - H_0: the first m autocorrelations are zero; $\hat{\rho}_k$: the sample autocorrelation at lag k
  - here $m = 10$, so the 5% critical value is 18.3
- Next chart
  - left, quantiles of the standardised returns against those of the Normal distribution
  - right, the ACF (autocorrelation function) of r_t and of $|r_t|$

Step 1: Stylised Facts of the BET



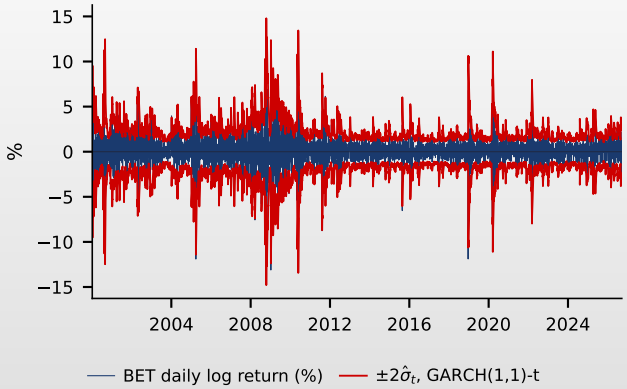
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Step 1: Reading the Chart

- **Heavy tails** (left)
  - Hill tail index of losses $\hat{\alpha} = 2.39$, 95% CI [2.12, 3.03]
  - the kurtosis is not reported, because $\hat{\alpha} < 4$
  - 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution
  - worst daily loss -13.1%, on 7 Jan 2009
- **Volatility clustering** (right)
  - ACF of $|r_t|$ at lag 1: 0.40, slow decay (the band assumes i.i.d. returns)
  - Ljung–Box $Q(10)$ on r_t^2: 2 440, against the critical value 18.3
  - the test on r_t^2 requires a finite eighth moment for the usual χ^2 limit
- First-order autocorrelation of the returns: 0.11
  - typical of a less liquid market (Chapters 2, 10)
- A Normal, constant-volatility VaR is therefore too low in stress and too high in calm periods

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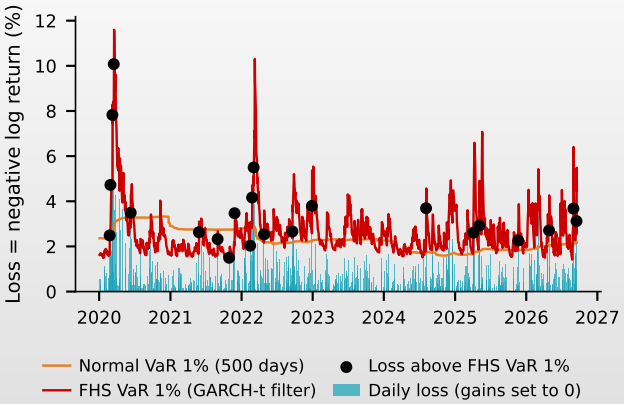
Step 2: GARCH(1,1)-t Volatility



- BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student-t innovations, 2000–2026
  - Robust standard errors: $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$
  - heavy tails even after filtering
- Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$
  - a half-life of about 74 trading days
  - it cannot be bounded away from 1 (pitfall "Persistence Near 1")
- Peak daily volatility 7.39% on 15 Oct 2008
  - at the end of the sample (18 Sep 2026), $\hat{\sigma}_T = 1.67\%$

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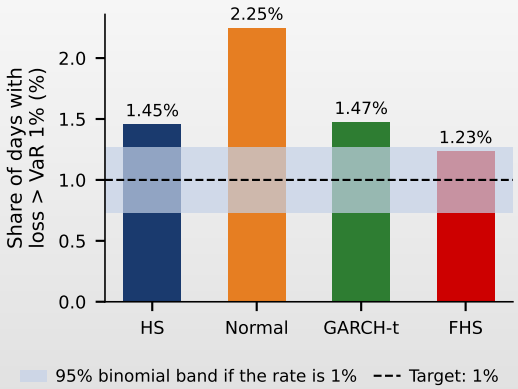
Step 3: Losses and VaR 1%, 2020–2026



- Daily BET losses (gains set to 0), Normal and FHS VaR 1%
  - black dots: losses above the FHS VaR
- FHS reacts within days
  - it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
  - The Normal VaR on a 500-day window moves only slowly
- 1 675 days, about 17 breaches expected
  - FHS 22, Normal 41
  - Normal breaches cluster in the first weeks of every stress episode
- Last VaR forecast: FHS 4.48%, Normal 2.18%

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Step 4: Breach Rates



- Share of days with a loss above the VaR 1% forecast, over the $T = 5\,432$ evaluation days of Step 3
- Light-blue band: the 95% binomial range [0.74%, 1.27%] if the true rate is 1%
- Only FHS (1.23%) lies inside the band
  - the Normal model (2.25%) breaches more than twice as often as it should

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Step 5: Results

Model	DQ	p_{DQ}	FZ0	p_{MCS}	ES zone
HS	192.1	<0.001	1.422	0.00	standard
Normal	364.6	<0.001	1.495	0.00	red
GARCH-t	17.9	0.007	1.175	0.18	green
FHS	9.9	0.129	1.157	1.00	green

- Only FHS passes the DQ test at 5% ($p = 0.129$)
  - GARCH-t fails it ($p = 0.007$) through its excess of breaches
- FZ0 (lower is better): FHS and GARCH-t are not significantly different (DM $t = 1.39$)
  - both beat HS and Normal ($|t| > 4.9$)
  - The 10% model confidence set is {GARCH-t, FHS}
- Comparative ES backtest: Normal red, GARCH-t and FHS green
- The GARCH-filtered models are superior
  - FHS remains the only model rejected by no test (coverage, independence, DQ)
- What the result supports: comparison of procedures
  - FZ0 scores the (VaR 2.5%, ES 2.5%) pair, not the VaR 1% of Steps 3–4

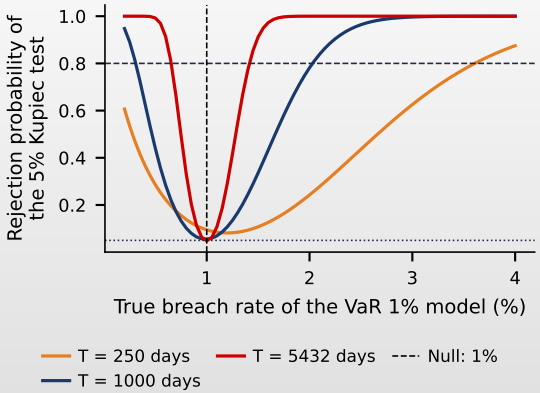
The Power of a Backtest: Definition

- The question: if the VaR 1% model is wrong, how often does the Kupiec test notice?
- Power at the true breach rate π: an exact binomial probability
  - the probability of the rejection region R_T of the 5% test

$$\text{Power}(\pi) = \sum_{x \in R_T} \binom{T}{x} \pi^x (1-\pi)^{T-x}, \qquad R_T = \{x : LR_{uc}(x) > 3.84\}$$

  - At $\pi = 1\%$ the same sum gives the actual size of the nominal 5% test: 9.48% at $T = 250$
  - At $T = 250$, R_T holds 0 breaches and any number from 7 up
- Minimum detectable effect (MDE): the smallest true rate $\pi > 1\%$ with $\text{Power}(\pi) \geq 80\%$
- Power grows with T
  - the standard error of the observed rate $\hat{\pi}$ is $\sqrt{\pi(1-\pi)/T}$

How Much Can a Backtest Detect? The Power of Kupiec



- Exact binomial probability that the 5% Kupiec test rejects a VaR 1% model
  - by its true breach rate
- $T = 250$ (one Basel year): a true rate of 2% is detected only 24% of the time
- MDE, searched on a grid with steps of 0.05 percentage points
  - 3.65% at $T = 250$, 2.05% at $T = 1\,000$, 1.45% at $T = 5\,432$
  - Even at $T = 5\,432$, a true 1.3% is detected with probability 55% only
- A non-rejection with low power is "no evidence", not "no problem"
  - state the power before the test

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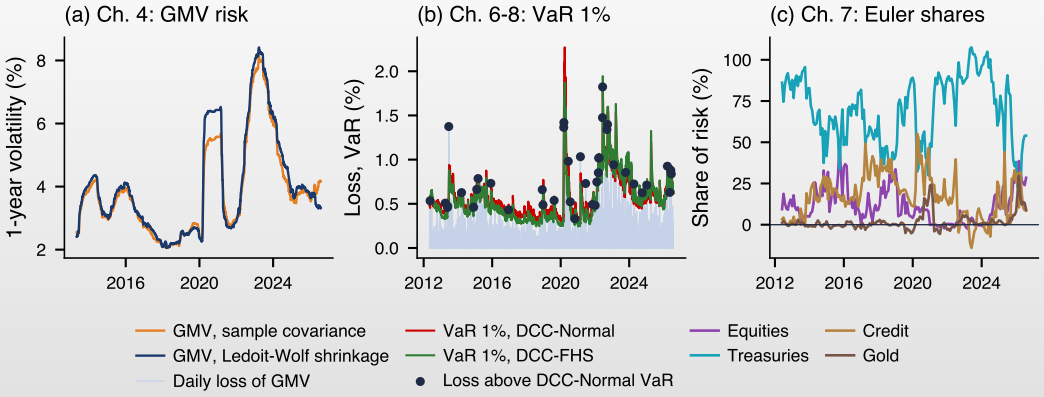
Multi-Asset Portfolio: the Question and the Decision Rule

- The question: which global minimum-variance (GMV) portfolio of eight ETFs do we hold, and which one-day VaR 1% forecast do we rely on?
- The multi-asset universe of Chapter 4, daily adjusted prices, Apr 2007 – Jul 2026
  - equities: SPY (US), EFA (developed markets ex-US), EEM (emerging markets)
  - US Treasuries: TLT (20+ years), IEF (7–10 years)
  - corporate bonds: LQD (investment grade), HYG (high yield); gold: GLD
- The evaluation: 1 May 2012 – 31 Jul 2026, 3 583 days
- Four chapters
  - Chapter 4: monthly GMV weights from the last 60 months, with the sample covariance or with Ledoit–Wolf shrinkage Ledoit & Wolf (2004)
  - Chapter 6: conditional covariance H_t from GARCH(1,1) and DCC Engle (2002), re-estimated every January
  - Chapter 7: portfolio VaR 1% and Euler risk contributions
  - Chapter 8: the Kupiec (coverage) and Christoffersen (independence, conditional coverage) tests

The decision rule

- Weights: the GMV variant with the lower volatility to 2018; from 2019 we test the choice
- Forecast: a VaR model rejected at 5% by none of the three tests
- On a tie: the model with the lower mean VaR

Multi-Asset Portfolio: Weights, VaR and Risk Contributions



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Multi-Asset Portfolio: the Steps in Numbers

- **(a) Chapter 4**: daily out-of-sample volatility, annualised (in the chart: over a rolling year)
  - GMV with the sample covariance: 4.13% a year
  - GMV with Ledoit–Wolf shrinkage: 4.28% a year, with a mean shrinkage intensity $\delta = 0.17$
  - monthly turnover: 0.085 with Ledoit–Wolf, against 0.135
  - gross exposure $\sum_i |w_i|$ of the target weights: mean 1.80 with Ledoit–Wolf, against 2.42; maximum 2.30, against 3.78
  - to 2018: 3.08% against 3.15%: the rule picks the sample covariance; from 2019: 4.87% against 5.07%
- **(b) Chapters 6–8**: $\sigma_{p,t} = \sqrt{w_t' H_t w_t}$, the forecast for day t, with w_t the weights at the start of the day (rebalanced monthly, then drifting with the prices)
  - DCC at the last re-estimation (Jan 2026, 4 712 days): $\hat{a} = 0.024$, $\hat{b} = 0.968$
  - DCC-Normal: VaR $= 2.326\,\sigma_{p,t}$
  - DCC-FHS: $\sigma_{p,t}$ times the 99% quantile of the standardised loss $-r_{p,s}/\sigma_{p,s}$ over the last 1,000 days
  - HS: the loss quantile over the last 500 days, with the weights w_t
  - 35.8 breaches of the VaR 1% expected
- **(c) Chapter 7**: the Euler share $w_i(H_t w_t)_i / w_t' H_t w_t$ of each asset class in the variance
  - it is also the share in VaR and ES under an elliptical distribution
  - mean over the evaluation: Treasuries 69%, credit 17%, equities 12%, gold 2%
  - 27 Mar 2020, the day with the highest $\sigma_{p,t}$ (0.98%): credit carries 63% of the risk with a weight of 25%

Multi-Asset Portfolio: the Result and the Decision

VaR 1%	Breaches	Rate	Kupiec p	indep. p	cond. p	mean VaR
DCC-Normal	40	1.12%	0.492	0.083	0.175	0.58%
DCC-FHS	52	1.45%	0.011	0.046	0.005	0.55%
HS, 500 days	48	1.34%	0.052	<0.001	<0.001	0.63%

- What the result supports: a retrospective comparison of procedures, on one portfolio and one period
  - only DCC-Normal passes all three tests; the VaR model is chosen on the same period on which it is evaluated
  - HS fails on independence, DCC-FHS on coverage and on independence
  - the DCC-FHS history applies today's weights to the data they were estimated on (the in-sample understatement of risk of Chapter 4)
  - standard deviation of $r_{p,s}/\sigma_{p,s}$: 0.91 inside the FHS window, 0.95 outside it
- Limits
  - the independence p-value of DCC-Normal is 0.083, below 10%
  - the Kupiec MDE at $T = 3\,583$: 1.55%; no transaction costs
- **What I choose**: GMV with the sample covariance and the DCC-Normal VaR 1% (0.64% for 31 Jul 2026)
- **Under which conditions**: monthly weights and DCC re-estimated yearly, with only the information available on the forecast date
- **What would make me drop it**
  - a rejection of independence on new data
  - a gross-exposure limit c: imposed in the optimisation at every rebalancing and checked daily; GMV breaches it for $c < 3.78$ (daily: 3.86), Ledoit–Wolf for $c < 2.30$ (2.34)

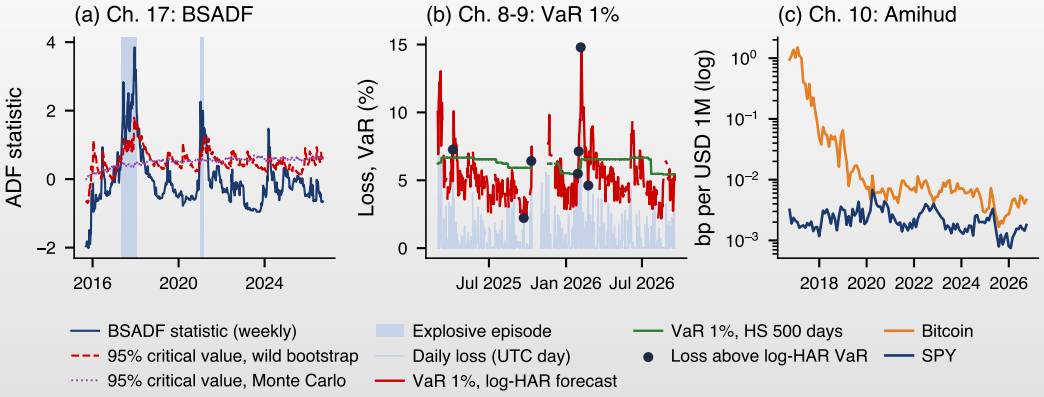
Bitcoin: the Question and the Decision Rule

- The question: a risk manager holds Bitcoin; which one-day loss limit does she set, and which signal would make her cut the position?
- Data
  - daily closes, 18 Sep 2014 – 18 Sep 2026: 4 384 returns
  - 5-minute bars on UTC (Coordinated Universal Time) days, 1 Sep 2024 – 17 Sep 2026: 659 complete days
- Five chapters
  - Chapter 1: stylised facts
  - Chapter 9: realised variance RV and log-HAR forecasts Corsi (2009)
  - Chapter 8: backtesting the VaR 1%
  - Chapter 10: spreads and Amihud illiquidity
  - Chapter 17: the GSADF bubble test and BSADF date-stamping Phillips, Shi & Yu (2015)

The decision rule

- Daily limit: the log-HAR VaR 1%, if it is not rejected at 5% and needs less capital on average than HS; otherwise the HS VaR
- On days without a log-HAR forecast (incomplete 5-minute bars): the HS VaR
- Cutting the position: BSADF above its 95% wild-bootstrap critical value for at least 7 weeks in a row
  - the minimum episode length of Chapter 17: $\lceil \ln T \rceil = 7$ weeks

Bitcoin: Bubbles, Realised-Volatility VaR and Liquidity



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Bitcoin: the Steps in Numbers

- Chapter 1: daily returns
  - Hill tail index of losses: 2.71 ($k = 109$, 2.5% of the days)
  - annualised volatility: 66.6%
  - ACF of $|r_t|$ at lag 1: 0.21
  - 29 days beyond ± 4 standard deviations, against 0.28 under the Normal distribution
- (b) Chapter 9: log-HAR on $\ln RV$, with the 1-, 7- and 30-day means, $R^2 = 0.36$
  - coefficients, Newey–West errors in brackets: $\hat{\beta}_d = 0.38$ (0.04), $\hat{\beta}_w = 0.47$ (0.08), $\hat{\beta}_m = -0.02$ (0.10)
  - expanding-window forecasts from 1 Mar 2025: 422 days, the same as in Chapter 9
- (b) Chapter 8: VaR 1% $= 2.326\sqrt{\widehat{RV}_t}$, against HS on the last 500 days
  - log-HAR VaR: mean 5.29%, between 1.99% and 14.45%; HS VaR: mean 6.24%
  - 144 of the 566 calendar days have no log-HAR forecast: the evaluation covers the days selected by data availability
  - $r_t/\sqrt{\widehat{RV}_t}$: standard deviation 1.00, excess kurtosis 2.54
  - 1% quantile of $r_t/\sqrt{\widehat{RV}_t}$: -2.52, against -2.33 for the Normal distribution
- (c) Chapter 10: Amihud illiquidity, in bp per USD 1 million traded
  - last two years: 0.0037 for Bitcoin, 0.0016 for SPY
  - monthly mean: 0.80 over the first 12 months from 2016, 0.0040 over the last 12 months
  - spreads from daily data disagree: Roll 85 bp, Corwin–Schultz 81 bp, EDGE (Efficient Discrete Generalized Estimator) -8 bp
- (a) Chapter 17: GSADF 3.85, against a 95% wild-bootstrap critical value of 2.82 (homoskedastic Monte Carlo: 2.27)
  - explosive episodes, with the wild-bootstrap critical values: May 2017 – Jan 2018 and Dec 2020 – Feb 2021
  - since 2022: BSADF above the wild critical value in 6 weeks, at most 3 in a row (peak 1.47, on 8 Mar 2024)
  - last value: -0.65, against a wild critical value of 0.58

Bitcoin: the Result and the Decision

VaR 1%	Breaches	Expected	Kupiec p	indep. p	cond. p	mean VaR
log-HAR	7	4.2	0.214	0.604	0.404	5.29%
HS, 500 days	7	4.2	0.214	0.604	0.404	6.24%

- What the result supports: a retrospective comparison of procedures, on 422 days with complete bars (1 Mar 2025 – 17 Sep 2026)
  - the same coverage, with a mean VaR lower by 0.95 percentage points
  - indep. p: only the 372 pairs of consecutive calendar days
  - the full rule (HS on days without a forecast), over all 566 days: 9 breaches, against 5.7 expected; Kupiec p 0.194, indep. p 0.589
  - the power is low: MDE 2.70%; a true rate of 2% is detected with probability 47% (power of the Kupiec test)
- Limits
  - only two years of 5-minute bars
  - the Normal quantile understates the tail: -2.52, against -2.33
  - Amihud uses the volume reported by aggregated exchanges
- **What I choose**: a daily limit equal to the log-HAR VaR 1% (5.04% for 17 Sep 2026), recomputed every day
- **Under which conditions**: the 5-minute bars cover at least 95% of the day; illiquidity stays of the order of that of SPY
- **What would make me drop it**
  - 5 or more breaches in 250 days (the Basel yellow zone, Chapter 8): I switch to the empirical quantile of $r_t / \sqrt{\widehat{RV}_t}$
  - BSADF above the wild critical value for 7 weeks in a row: I cut the position
  - weekly monitoring repeats the test: a pointwise 95% threshold does not control false alarms over the whole horizon, so I confirm with the family-wise threshold of Chapter 17

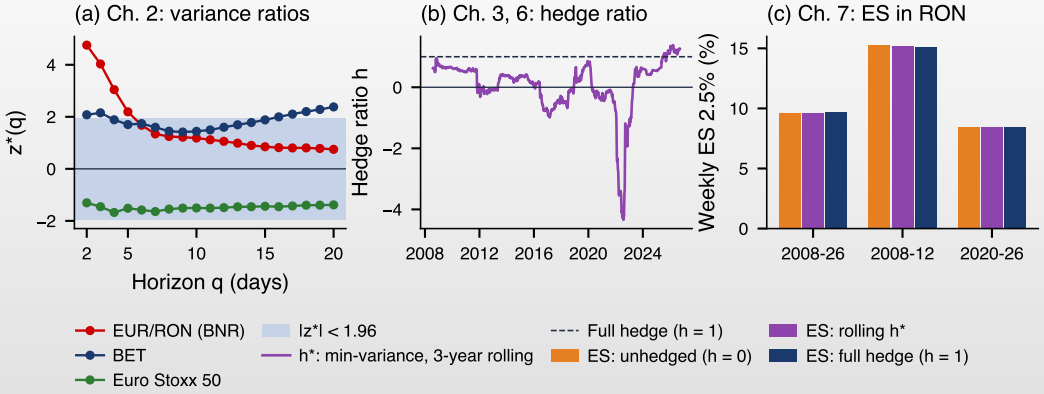
A Romanian Investor: the Question and the Decision Rule

- The question: an investor who measures her wealth in lei holds 50% BET and 50% Euro Stoxx 50, bought in euro; does she hedge the EUR/RON (lei per euro) currency risk of the euro leg?
- Data: BET and Euro Stoxx 50 (daily closes), the BNR EUR/RON reference rate, Jul 2005 – Sep 2026
  - exposures and hedging on weekly (Friday) returns: the BNR fixes the rate at 13:00 and the exchanges close at different times
- Five chapters
  - Chapter 2: variance ratios Lo & MacKinlay (1988)
  - Chapter 3: exposures with Newey–West errors
  - Chapter 5: GARCH(1,1)-t volatility
  - Chapter 6: the minimum-variance hedge ratio h^* over a rolling three years
  - Chapter 7: VaR 1% and ES 2.5% by historical simulation
- The benchmark Campbell, Serfaty-de Medeiros & Viceira (2010): a currency that is negatively correlated with equities lowers risk, so a full hedge does not minimise risk

The decision rule

- Three choices, with h the share of the euro leg sold forward: $h = 0$ (no hedge), rolling h^*, $h = 1$ (full hedge)
- We pick the choice with the lowest weekly ES 2.5%
- If the ES values differ by less than 0.25 percentage points in every period: $h = 0$, which needs no forward contracts

A Romanian Investor: Efficiency, Currency Hedging and ES in Lei



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A Romanian Investor: the Steps in Numbers

- **(a) Chapter 2**: the robust statistic $z^*(q)$ on daily returns; $|z^*| > 1.96$ rejects the random walk at 5%
  - EUR/RON: $z^*(2) = 4.76$, with VR(2) $= 1.17$ and a first-order autocorrelation of 0.17
  - BET: $z^*(2) = 2.08$; the Chow–Denning test over $q = 2, 5, 10, 20$: $p = 0.067$
  - Euro Stoxx 50: $|z^*(q)| \leq 1.68$; Chow–Denning: $p = 0.43$
  - the daily EUR/RON risk times $\sqrt{h}$ understates the h-day risk, so we work on weeks
- **Chapter 3**: exposures over 1 101 weeks, with $f = \Delta$EUR/RON ($f > 0$: the leu depreciates)
  - Euro Stoxx 50 in euro on f: $\hat{\beta} = -0.52$ ($t = -2.61$)
  - BET on Euro Stoxx 50 and f: 0.56 ($t = 7.4$) and -0.54 ($t = -2.11$), $R^2 = 0.30$
  - the leu depreciates in the weeks in which equities fall
- **Chapter 5**: GARCH(1,1)-t on daily returns
  - EUR/RON: persistence 1.000, at the boundary of stationarity
  - EUR/RON: volatility today 0.96% a year, against a mean of 4.4%
  - BET: 25.9% today, against a mean of 21.2%
  - Euro Stoxx 50: 13.8% today, against a mean of 21.0%
- **(b) Chapter 6**: $h^* = \mathrm{Cov}(r^{RON}, f)/\mathrm{Var}(f)$, with $r^{RON} = r^{EUR} + f$ the return in lei of the euro leg
  - full sample: $h^* = 0.48$
  - over a rolling three years: between -4.34 (Jul 2022) and 1.38 (Mar 2026); last value 1.27
  - the three-year correlation between the Euro Stoxx 50 and f is negative in 93% of the windows
- **(c) Chapter 7**: weekly ES 2.5% of the 50/50 portfolio, rebalanced weekly, for the three choices (table on the next slide)
  - the exact simple return $R_p = \frac{1}{2}(e^b - 1) + \frac{1}{2}[e^{s+f} - 1 - h(e^f - 1)]$, with b, s the log returns of the BET and the Euro Stoxx 50; a forward on the notional at the start of the week, with zero carry
  - the loss, in log points: $L = -100 \ln(1 + R_p)$

A Romanian Investor: the Result and the Decision

Weekly ES 2.5%	Weeks	$h = 0$	rolling h^*	$h = 1$
Aug 2008 – Sep 2026	944	9.60%	9.62%	9.67%
Aug 2008 – Dec 2012	228	15.23%	15.20%	15.12%
Jan 2020 – Sep 2026	351	8.43%	8.43%	8.47%

- The retrospective conclusion: the ES values of the three choices differ by at most 0.11 percentage points, so the choice of h mattered little for the risk
  - annualised weekly volatility: EUR/RON 4.6%, Euro Stoxx 50 20.9%
  - the negative correlation lets the depreciation of the leu offset part of the losses of the euro leg
- What the result supports: comparison of hedging choices, for one portfolio
- Limits
  - the cost of the hedge, the RON–EUR interest differential, is not in the course data
  - the depreciation of the leu added 1.55% a year to the euro leg (Jul 2005 – Sep 2026); a forward replaces it with that differential
  - h^* is unstable when $\mathrm{Var}(f)$ is small, as in 2019–2022: the estimation risk of Chapter 4
  - h^* hedges the euro leg; for the whole portfolio, $\mathrm{Cov}(b + s + f, f)/\mathrm{Var}(f) = -0.35$ (linear approximation)
- **What I choose now**: the full hedge ($h = 1$)
  - the three-year correlation is positive after 12 Sep 2025 (0.037 at the end): the rule for dropping $h = 0$ was triggered on 13 Mar 2026
  - the rolling h^* (1.27) would sell forward more than the euro leg
- **Under which conditions**: the three-year correlation between equities and EUR/RON stays positive
- **What would make me drop it**: a negative three-year correlation for six months in a row (back to $h = 0$)

The Project in Four Steps

- An empirical study of one question about financial markets, answered with the methods of this course
  - **70%** of the final grade
  - quizzes and seminar activity 20%, attendance 10%, no final exam
- **1. Replicate**: one full table or figure of a published paper (or of the lecture)
  - cell by cell (standard: next slide)
- **2. Extend**: new data, a new market (the BVB, crypto) or a more recent sample
  - under a **pre-analysis plan** recorded in a dated commit before the test period is opened
  - The plan states the null, the test, the loss, the sample split, the identification strategy, the power of the main test and the specification grid with its correction
- **3. Deliver on GitHub**: a repository that regenerates every number
  - with `AI_USE.md` and `AI_ERRORS.md` mandatory
- **4. Present and defend**: team presentation, then a 10-minute oral defence
  - Each member explains one result and answers a "what changes if..." question
  - Grades can differ between members

Replication Standard

- Target: one full table or figure, with its DOI, page and table number, not a single coefficient
  - Report a cell-by-cell comparison: published value, your value, difference, and the reason for each gap (sample, filter, data vendor, definition)
- Large-scale replications show why
  - many anomalies shrink out of sample or with better data Hou, Xue & Zhang (2020)
  - most factors replicate once the design is right Jensen, Kelly & Pedersen (2023)
- Reproducibility
  - a lockfile of package versions, fixed random seeds, series names and dates of the data
  - one command that regenerates every number
  - Different teams on the same data disagree more than their standard errors suggest Menkveld et al. (2024)
  - more reproducible work is associated with a smaller gap (research case study, this chapter)

Choosing the Question

A good question is

- **Specific**: one market, one period, one quantity
- **Testable**: a null hypothesis and a benchmark
- **Feasible**: data you can obtain and clean in a few weeks
  - the course data sets in the course GitHub repository
  - the National Bank of Romania, BNR; Federal Reserve Economic Data, FRED; Kenneth French Data Library; DefiLlama
- **Relevant**: an answer matters to an investor, a bank or a regulator

Examples

- Too vague: "Is Bitcoin risky?"
  - Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin?"
  - Then: "Which model forecasts it best since 2024?"
- Too broad: "Can machine learning predict the market?"
  - Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Project Grading Criteria

Component	Grading criteria
Replication (step 1)	A published table or figure reproduced cell by cell; every gap explained
Question, data, extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Plan, identification, power, real-time design, benchmark, specification curve with a correction
Interpretation and report	Economic meaning, limitations, honest conclusion; 8–12 pages, 10–15 references
GitHub repository (step 3)	One command reproduces every result (lockfile, seeds); `AI_USE.md`, `AI_ERRORS.md`
Presentation and defence (step 4)	Individual: one result explained, answers to "what changes if..."

Deliverables and Timeline

- Week 1 after this chapter: team, question, one-page proposal
  - proposal: Seminar 19, C2
  - GitHub repository
- Weeks 2–3: data, replication (step 1)
  - extension plan recorded in a dated commit
- Weeks 4–5: extension, evaluation, robustness (step 2)
- Final weeks: the 8–12-page report
  - repository with README and the two AI files (step 3)
  - presentation and oral defence (step 4)
- Exact dates are announced at the course and seminar

The AI Policy and the Two Mandatory Files

Policy

- **Allowed**: AI assistants for code, debugging, literature search and writing
- **Declared**: undeclared use of AI counts as plagiarism
- **Checked**: you are responsible for every line
  - a reference that does not exist counts as fabricated data
- **No AI during the oral defence**: "the AI wrote it" is not an explanation

The two files

- `AI_USE.md`: tools and versions
  - who used them for what
  - where AI-written code or text was kept
  - how you verified the replicated table, data, formulas and references
- `AI_ERRORS.md`: at least **three** places where the assistant was wrong
  - what you asked, what it answered, what was wrong, how you found out, the correct version
- Errors you caught are a strength of the project, not a weakness

How Much Do Researchers' Choices Matter? (1/2)

- Menkveld et al. (2024): 164 research teams test the same six hypotheses on the same data
  - *Journal of Finance* 79(3)
  - Finance Crowd Analysis Project, #fincap
- **Standard error**
  - Uncertainty from sampling, the data-generating process
  - **Nonstandard error**: uncertainty from the analysis path each researcher picks, the evidence-generating process
  - Model: team j reports $X_j = e_j + \varepsilon_j$, e_j the researcher fixed effect of its path, ε_j the sampling error
  - Nonstandard error $:= Q_{0.75}(x) - Q_{0.25}(x)$, the interquartile range (IQR) of the estimates across teams, $Q_p(x)$ being the p-quantile of the team estimates x
  - The IQR, not the standard deviation, because the e_j can have fat tails

How Much Do Researchers' Choices Matter? (2/2)

- **Data**
  - All trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records
  - Six hypotheses of "no annual trend" (market efficiency, realised spread, client share of volume, ...)
  - Each team reports the average yearly change in % with its standard error (SE)
  - Four stages in 2021: independent analysis, peer feedback, the five best papers shared, final estimates
  - design details in the Appendix
- Link to this chapter: every team project is one path through the "garden of forking paths"
  - Pre-registration, the specification grid and a repository that reruns every number are the remedies

Key Results: Size and Quality

- **Size**
  - For market efficiency the median estimate is -1.1% per year, with an IQR of 6.7 percentage points (pp)
  - Even for the client share of volume, a simple ratio, the IQR is 1.2 pp around -3.3% (Table I, Panel C)
  - The median SE of 2.5% implies a Gaussian IQR of $1.35 \times 2.5 = 3.4\%$, half the 6.7 pp
  - The no-dispersion null is rejected for all six hypotheses, $p < 0.0001$ (Table II)
- **Quality** (Table III, Fig. 2)
  - a rise of one standard deviation in reproducibility is associated with a 25.0% lower IQR
  - the same rise in peer rating, with a 33.3% lower IQR
  - Code that reruns end to end is associated with answers closer to the consensus
- What the result supports: description
  - of the dispersion across teams; the link with quality is only an association

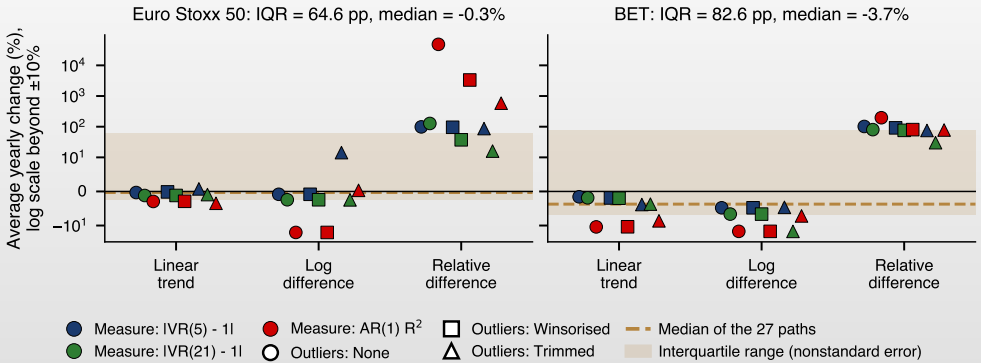
A Mini-Multiverse on Course Data: the Design

- Daily closes 2002–2018 from the course data, without the artificial non-trading-day fills (the rule of Chapter 2)
  - Forks: outliers × measure × model, 27 paths
  - The band is the IQR, the nonstandard error of the multiverse
- Measures of inefficiency
  - $|VR(q) - 1|$ ($VR(q)$: the variance ratio of Chapter 2, with the Lo–MacKinlay corrections, $q = 5, 21$ days) and the R^2 of an autoregressive AR(1) model: departures from a random walk
  - A positive trend means declining efficiency

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A Mini-Multiverse on Course Data: 27 Paths per Index

Legend: *VR*(*q*), the *q*-day variance ratio; AR(1), the first-order autoregressive model; IQR, the interquartile range; pp, percentage points.



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A Mini-Multiverse on Course Data: Results

- IQR of the estimates: 64.6 pp (Euro Stoxx 50) and 82.6 pp (BET)
  - around medians of -0.3% and -3.7% a year
- The model fork creates the dispersion
  - Over the nine linear-trend paths the IQR is only 2.6 pp for the Euro Stoxx 50 and 6.7 pp for the BET
  - Relative differences explode: medians $+98.7\%$ (Euro Stoxx 50) and $+80.3\%$ (BET), up to 48 415%
- Sampling benchmark: the Gaussian IQR, $1.35\times$ the median SE
  - 41.3 pp (Euro Stoxx 50) and 43.4 pp (BET)
  - The IQR exceeds this benchmark for both indices
- What the result supports: description
  - of the dispersion over the 27 paths of the grid, not across real teams

Limits and an Open Extension

- **Authors**: the multiverse covers a few forks only and explains the reported estimates poorly
- **Authors**: beliefs were asked as standard deviations, not IQRs
- **Authors**: the test of dispersion assumes consistent reported SEs
- **Our replication**: a multiverse spans only the options in its grid
  - so its IQR neither estimates nor bounds from below the dispersion across real teams
- **Our replication**: daily closes only
  - so the paper's intraday frequency pairs cannot be run
- **Open extension**: does a pre-registered grid shrink the nonstandard error of a risk question?
  - Measure the IQR of VaR 1% conclusions for the BET across defensible paths

An Open Question

- **Open question**: how much of a result is the data, and how much the researcher?
  - Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries Jensen, Kelly & Pedersen (2023)
  - AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
  - Formal version: the conclusion must hold over a pre-registered set $\mathcal{S}$ of defensible specifications, with the family corrected (FWER or FDR)
  - Report its distribution over $\mathcal{S}$, not the best cell

The Discovery Loop and Its Checks (1/2)

- **Loop**
  - Question and H_0/H_1 (week 1)
  - Literature and the replication target seen in the PDF (weeks 1–2)
  - Pre-registration of the extension, recorded in a dated commit before the first extension result (week 2)
  - Replication as a passing test in the repository (weeks 2–3)
  - The whole grid (weeks 4–5)
  - An AI examiner before the defence

The Discovery Loop and Its Checks (2/2)

- **Checks by the researcher**
  - The target number is in the paper
  - The plan precedes the first results file in the commit history
  - Every run is logged and counted
  - A fresh clone regenerates every number
  - Each member can derive the key formula without AI
  - Prompts for every step and the test for each check: Appendix

Mini-Case: Replication and Extension (1/2)

- **Replicate**: FHS VaR 1% for the BET
  - Chapter 8 reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
  - The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$)
  - Same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal: "only GARCH-EVT is correctly calibrated on both markets"
  - the assistant reads each p-value alone at 5%

Mini-Case: Replication and Extension (2/2)

- **Extend**
  - The 12 conditional-coverage tests of Chapter 8 (S&P 500 and BET $\times$ 6 models) form one family
  - Bitcoin and EUR/RON stay outside the family: on these markets some GARCH forecasts are missing (Chapter 8)
  - Rejections at 5%: 8 without correction, 6 with Holm (FWER), 7 with Benjamini–Hochberg, BH (FDR 5%)
  - FHS fails only on the S&P 500 ($p = 0.040$); within the family of 12 this rejection is no longer significant, so after Holm FHS and GARCH-EVT pass on both markets
- Interpretation: "FHS fails for the S&P 500" depends on whether the test is read alone or in its family
- What the result supports: comparison of procedures
- In the project, fix the family in the plan before seeing results and report both readings

Your Turn: the Capstone Plan

- **Question**: your project idea from Part C, written as H_0/H_1, with the observation that would falsify it
- **Replicate first**: one published table seen in the paper, turned into a passing test in the repository
  - Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- **Deliverables** along the timeline
  - proposal and replication target, pre-registration commit
  - passing replication test, grid of results
  - mock defence, report, oral defence
- **AI log** (templates are in the project folder)
  - the prompts that changed the design, in `AI_USE.md`
  - every caught error, with the test that caught it, in `AI_ERRORS.md`

Key Takeaways

- Returns are hard to predict, their risk is not
  - volatility, tails and dependence can be forecast
- Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- Heavy tails change inference
  - use robust errors and the bootstrap
  - report intervals, state the power
- Most wrong results come from the design, not from the model
  - look-ahead, overfitting, multiple testing, calendars
- The analysis path is itself a source of uncertainty
  - pre-register, replicate, report the whole grid
- New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Check Your Understanding (1/2)

Questions

- ⊡ Why does the Kupiec test need $\alpha = 0.01$, not 2.39 or 0.192?
- ⊡ Why is the sample excess kurtosis of BET returns not an estimate of anything?
- ⊡ Why is the Wald interval for GARCH persistence uninformative once we restrict to $p \leq 1$, and what replaces it?
- ⊡ What does a non-rejection of the Kupiec test at $T = 250$ tell you?
- ⊡ Why is the IQR of a 27-path multiverse not the nonstandard error of 164 teams?

Check Your Understanding (2/2)

Further study

- Seminar 19: the exercises A1–A12, B1–B8 and C1–C3 apply this chapter to the BET
- Textbooks: Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)
- Advanced courses on quantinar.com: Statistics of Financial Markets
  - Measuring Statistical Risk
- Code and data: Quantlets/Ch_19 (GitHub)

Two Computations on New Numbers (1/2)

Try it

- **1.** A VaR 1% model has $x = 6$ breaches in $T = 400$ days
  - Compute LR_{uc} with the Kupiec formula of the Chapter 8 review and decide at 5%
- **2.** A GARCH(1,1) has $\alpha + \beta = 0.97$
  - Compute the half-life $h_{1/2}$
  - How does it change if $\alpha + \beta = 0.985$?

Two Computations on New Numbers (2/2)

Check your answer

1. $\hat{\pi} = 6/400 = 1.5\%$

$$LR_{uc} = 2\left[(394 \ln 0.985 + 6 \ln 0.015) - (394 \ln 0.99 + 6 \ln 0.01)\right] = 0.88$$

- Below 3.84: not rejected
- At $T = 400$ the test has little power, so this is "no evidence", not "no problem"

2. $\ln 0.5 / \ln 0.97 = 22.8$ days and $\ln 0.5 / \ln 0.985 = 45.9$ days

- Halving $1 - \alpha - \beta$ roughly doubles the half-life

The Practical Decision

- **What I choose**: for a risk project, a GARCH filter with a flexible tail (FHS), against historical simulation as the simple benchmark
  - on the BET, FHS is the only model rejected by no test (coverage, independence, DQ)
  - the 10% model confidence set on FZ0 is {GARCH-t, FHS}
- **Under which conditions**
  - real-time forecasts, with the information set $\mathcal{F}_{t-1}$ and parameters re-estimated on a rolling window
  - an evaluation with enough power: the Kupiec MDE is 1.45% at $T = 5\,432$, against 3.65% at $T = 250$
  - the loss function and the test family fixed in the plan by a dated commit
- **What result would make me drop it**
  - a rejection that survives the Holm correction over the family of conditional-coverage tests
  - a conclusion that changes across the pre-registered specification grid
  - a replication of the target table with deviations I cannot explain cell by cell

Step 2: the GARCH(1,1)-t Model

- The GARCH(1,1) of the Chapter 5 review, with Student-t innovations Bollerslev (1986)

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2$$

  - σ_t: conditional volatility, known at $t-1$
  - z_t: i.i.d. Student-t with ν degrees of freedom, scaled to variance 1
  - $\omega > 0$, $\alpha, \beta \geq 0$: α measures the reaction to news, β the memory
  - a small ν means heavy tails
- Persistence $\alpha + \beta$ and half-life $h_{1/2}$: the formulas of the Chapter 5 review
- Estimated by maximum likelihood, with robust (sandwich) standard errors

Step 2: Today's Volatility and the Long-Run Level (1/2)

- Two numbers at the end of the sample
  - $\hat{\sigma}_T$ is known at $T-1$
  - $\hat{\sigma}_{T+1}$ also uses the last return $r_T = 2.17\%$ and is the input of tomorrow's VaR

$$\hat{\sigma}^2_{T+1} = \hat{\omega} + \hat{\alpha}\,(r_T - \hat{\mu})^2 + \hat{\beta}\,\hat{\sigma}^2_T = 0.0448 + 0.192\,(2.17 - 0.08)^2 + 0.799 \times 1.67^2 = 3.11$$

  - So $\hat{\sigma}_{T+1} = \sqrt{3.11} = 1.76\%$: a large last return lifts the forecast above $\hat{\sigma}_T = 1.67\%$
- Long-run variance implied by the model

$$\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta} = \frac{0.0448}{0.0093} = 4.82$$

  - Returns are in %: $\bar{\sigma} = 2.20\%$ a day (34.7% a year), against a sample standard deviation of 1.40% a day (22.1% a year)

Step 2: Today's Volatility and the Long-Run Level (2/2)

- Why the gap is not a contradiction: the denominator $1 - \alpha - \beta = 0.0093$ is small and imprecise
  - Holding ω fixed, $p = 0.9867$ gives $\bar{\sigma}^2 = 3.37$ and $p = 0.9947$ gives 8.47
  - the profile CI reaches $p = 1$, where $\bar{\sigma}^2$ is infinite
  - Use $\hat{\sigma}_{T+1}$ and the short-horizon forecast path
  - Treat the long-run level as weakly identified
- The forecast path returns geometrically to $\bar{\sigma}^2$, at the rate $\hat{\alpha} + \hat{\beta}$

$$\hat{\sigma}^2_{T+h} = \bar{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^{h-1}(\hat{\sigma}^2_{T+1} - \bar{\sigma}^2), \qquad h = 21\colon\ 4.82 + 0.9907^{20}\,(3.11 - 4.82) = 3.40$$

  - So $\hat{\sigma}_{T+21} = 1.84\%$: after one month the forecast has covered only 17% of the distance to the long-run level

Step 3: Four VaR 1% Forecasts (1/2)

- Unconditional, on the last 500 days: one quantile for the whole window
  - **HS** (historical simulation): the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
  - **Normal**: $\mathrm{VaR} = -(\hat{\mu} + \hat{\sigma} z_{\mathbf{0.01}})$
  - $\hat{\mu}$, $\hat{\sigma}$: sample mean and standard deviation; $z_{\mathbf{0.01}} = -2.326$: the 1% quantile of the standard Normal distribution
- Conditional, GARCH-t on the last 1000 days
  - the quantile scales with today's volatility forecast $\hat{\sigma}_t$

$$\textbf{GARCH-}t\text{:}\ \ \mathrm{VaR}_t = -\left(\hat{\mu} + \hat{\sigma}_t\, t_\nu^{-\mathbf{1}}(0.01)\sqrt{(\nu-2)/\nu}\right), \qquad \textbf{FHS}\text{:}\ \ \mathrm{VaR}_t = -\left(\hat{\mu} + \hat{\sigma}_t\, \hat{q}_{\mathbf{0.01}}(z)\right)$$

  - $t_\nu^{-\mathbf{1}}(0.01)$: the 1% quantile of a Student-t with ν degrees of freedom
  - $\sqrt{(\nu-2)/\nu}$ rescales it to variance 1
  - **FHS** (filtered historical simulation): $\hat{q}_{\mathbf{0.01}}(z)$, the empirical 1% quantile of the standardised residuals $z_t = (r_t - \hat{\mu})/\hat{\sigma}_t$, replaces the Student-$t$ quantile McNeil & Frey (2000)

Step 3: Four VaR 1% Forecasts (2/2)

- Real-time design
  - Forecast for day t uses data up to $t-1$ only
  - GARCH re-estimated every 21 trading days
  - $\hat{\sigma}_t$ updated daily
- Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
  - A breach (hit) on day t: the loss $-r_t$ exceeds VaR_t
  - Expected breaches at 1%: $0.01\,T \approx 54$
- Unconditional models react only as the window rolls
  - conditional ones react through $\hat{\sigma}_t$, within days

Step 4: the Three Backtests

- Kupiec coverage test Kupiec (1995): is the breach rate equal to $\alpha = 1\%$?
  - LR_{uc} as in the Chapter 8 review, rejected at 5% when $LR_{uc} > 3.84$
- Christoffersen independence test Christoffersen (1998): is a breach today more likely after a breach yesterday?

$$LR_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}}\, \hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\, \hat{\pi}_{01}^{n_{01}}\, (1-\hat{\pi}_{11})^{n_{10}}\, \hat{\pi}_{11}^{n_{11}}} \;\sim\; \chi^2_1$$

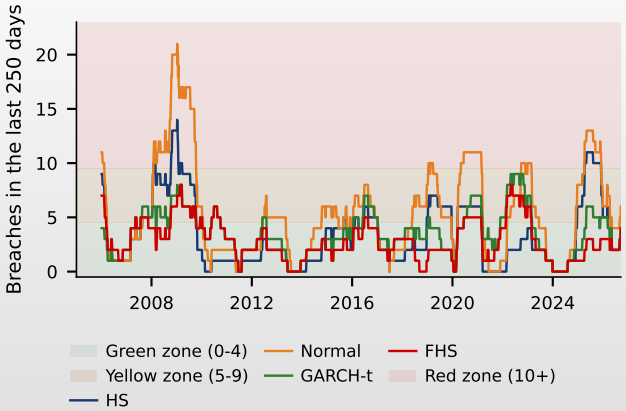
  - n_{ij}: number of days in state j after a day in state i (1 = breach, 0 = no breach)
  - $\hat{\pi}_{01} = n_{01}/(n_{00}+n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10}+n_{11})$: breach rates after a quiet day and after a breach
  - $\hat{\pi} = (n_{01}+n_{11})/(n_{00}+n_{01}+n_{10}+n_{11})$: the rate over the $T-1$ transitions, within $1/(T-1)$ of x/T
- Joint test: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2_2$, with 5% critical value 5.99

Step 4: Coverage and Independence Tests

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

- Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- Only FHS is rejected by none of the three tests at 5%
- HS and Normal also fail independence
  - their breaches cluster
- GARCH-t: independence is not rejected, but there are too many breaches
  - the Student-t tail is not heavy enough at 1%

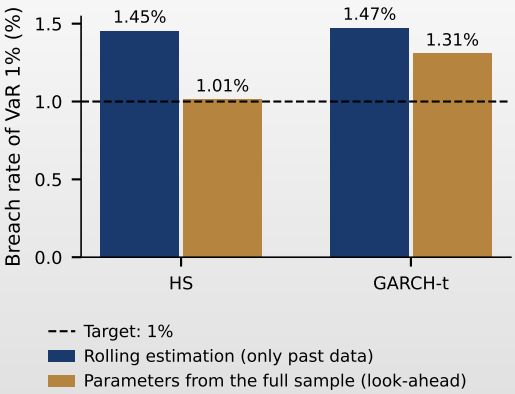
Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- Normal and HS enter the red zone in episodes, when their breaches cluster
  - shares: last column of the Step 4 table
- The two GARCH-filtered models never enter it

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Look-Ahead Bias in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction
- GARCH-t with full-sample parameters: 1.31% of days
  - with rolling estimation 1.47%
- Both look better than their real-time versions
  - the gap is the value of information nobody had
- Only the rolling result is evidence about a forecasting model

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Step 5: Three Tools for a Formal Comparison (1/2)

- **Dynamic Quantile (DQ)** test Engle & Manganelli (2004): are the breaches predictable?
  - Regress $I_t - 0.01$ ($I_t = 1$ if the loss exceeds VaR_t, else 0) on a constant, $I_{t-1}, \ldots, I_{t-4}$ and VaR_t
  - *DQ*, the Wald statistic that all six coefficients are zero, is χ^2_6 under a correct model
  - It detects breaches predictable from the past or from the level of VaR, which the Christoffersen test (one lag) can miss
- **FZ0 score** Patton, Ziegel & Chen (2019): one number per day that scores a (VaR, ES) forecast pair

$$S(v, e; y) = -\frac{1\{y \le v\}(v - y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$$

  - lower is better
  - $\alpha = 2.5\%$: the ES level; y: the realised return
  - $v = -\mathrm{VaR}$, $e = -\mathrm{ES}$: the forecasts written as return quantiles ($e \le v < 0$)
  - Strictly consistent for (VaR, ES) Fissler & Ziegel (2016): on average the true pair has the lowest score

Step 5: Three Tools for a Formal Comparison (2/2)

- **Pairwise Diebold–Mariano tests** Diebold & Mariano (1995): t_{DM} as in the Chapter 14 review
  - d_t: the FZ0 difference of the two models on day t
  - $t_{DM} \approx N(0,1)$ under equal accuracy
- **Model confidence set** at 10% Hansen, Lunde & Nason (2011)
  - it contains the best model with probability 90%
  - a model stays in it while $p_{MCS} \geq 0.10$
- **Comparative ES backtest** Nolde & Ziegel (2017), against HS as the standard model
  - red if the model is significantly worse
  - green if significantly better
  - yellow otherwise

Step 5: Which Forecast Pair Does FZ0 Score?

- A second forecast pair from the same four models: VaR 2.5% and ES 2.5%
  - at the Basel ES level $\alpha = 2.5\%$
  - Not the VaR 1% of Steps 3–4: FZ0 is consistent only when v and e are the quantile and the tail mean at the same level α
- Each model's ES 2.5% is the mean loss beyond its own VaR 2.5%
  - from the same loss distribution it uses for VaR in Step 3
  - **HS**: the estimator of Chapter 7; of the 500 losses, the 12 largest plus half of the 13th, divided by 12.5
  - **Normal**: $\mathrm{ES} = -\hat{\mu} + \hat{\sigma}\,\varphi(z_{\mathbf{0.025}})/0.025 = -\hat{\mu} + 2.338\,\hat{\sigma}$, with φ the standard Normal density
  - **GARCH**-t: $-\hat{\mu}$ plus $\hat{\sigma}_t$ times the upper-tail mean of the Student-t rescaled to variance 1
  - **FHS**: the negative of the sum of the conditional mean and $\hat{\sigma}_t$ times the mean of the 25 smallest of the 1,000 standardised residuals z_t (the same estimator)

Step 5: The ES 2.5% Forecasts in Numbers

Model	Average VaR 1% (Steps 3–4)	Average ES 2.5% (Step 5)
HS	3.66%	3.99%
Normal	2.90%	2.92%
GARCH-t	2.95%	3.08%
FHS	3.17%	3.41%

- Averages of the daily forecasts over the $T = 5\,432$ evaluation days, losses in % of the position
- Normal model: ES 2.5% / VaR 1% $= 2.338/2.326 \approx 1.005$
  - the ratio in the table (2.92/2.90)
  - an ES 1% would be about 15% higher
- The gap between the columns grows with the weight of the tail the model allows
  - about 9% for HS and 7.5% for FHS (empirical tails)
  - 4% for GARCH-t
  - 0.5% for the Normal

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability	Chapters 1–2: moments, robust tests, variance ratios
Expected returns, factors, portfolios	Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR
Risk measures and capital	Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with machine learning (ML), deep learning or text	Chapters 13–15: purged cross-validation (CV), foundation models, sentiment
Crypto, bubbles, systemic risk	Chapters 16–18: stablecoins, GSADF, CoVaR, networks

- Starting points: the Part C question of every seminar (table in the Appendix)
  - Your own question is welcome if it passes the five questions of this chapter

Identification: from Correlation to a Claim (1/2)

- Event questions need a counterfactual
  - e.g. "did spot ETFs, the FTSE Russell upgrade or a bank tax change X?"
  - A control market not hit by the event: difference-in-differences on the risk measure Y (e.g. volatility or VaR 1%)

$$\hat{\delta} = \left(\bar{Y}^{\text{after}}_{\text{treated}} - \bar{Y}^{\text{before}}_{\text{treated}}\right) - \left(\bar{Y}^{\text{after}}_{\text{control}} - \bar{Y}^{\text{before}}_{\text{control}}\right)$$

  - Equivalently: δ in $Y_{it} = a + b\,D_i + c\,P_t + \delta\,D_i P_t + \varepsilon_{it}$, with $D_i = 1$ for the treated market, $P_t = 1$ after the event and HAC standard errors
  - Identifying assumption: parallel trends (without the event Y would have moved alike in both markets), checked on the pre-event period

Identification: from Correlation to a Claim (2/2)

- A synthetic control built from several untreated markets Abadie, Diamond & Hainmueller (2010)
- Placebo event dates: the event moved one year earlier or later
- Unknown timing: a sup-Wald break test Andrews (1993)
  - report the estimated break date, not an imposed one
- Forecasting questions: out of sample only, with a test period declared in the plan
- One event is $n = 1$
  - say what the design can and cannot identify

Power and Robustness without p-Hacking (1/2)

- **Power**: before the main test, compute the minimum detectable effect (MDE)
  - MDE: the smallest effect the design detects with 80% power at 5%
  - For an approximately Normal estimate with standard error SE, a two-sided 5% test and 80% power

$$\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\mathrm{SE} = 2.80\,\mathrm{SE}, \qquad T_{\min} = \left(\frac{2.80\,\sigma}{\delta}\right)^2 \text{ when } \mathrm{SE} = \sigma/\sqrt{T}$$

  - z_p: the p-quantile of the standard Normal distribution; δ: the effect you look for; σ: the standard deviation of one observation
  - Deliverable: a simulated power curve of the main test at your T
  - Short crypto, ETF or post-event samples make many questions underpowered (Kupiec at $T = 250$: MDE 3.65%)

Power and Robustness without p-Hacking (2/2)

- Specification curve
  - estimate every defensible combination of window, lags, α and sample start
  - Plot all estimates sorted, with their intervals Simonsohn, Simmons & Nelson (2020)
  - Report the spread across specifications, not the best cell
- Correct the family
  - Holm for any false discovery
  - Benjamini–Hochberg for their share
  - or White's bootstrap reality check for data snooping White (2000)
  - Significance thresholds must rise with the number of tests tried Harvey & Liu (2020), Harvey (2017)

Forecast Evaluation for Risk Projects

- Volatility: a robust loss (MSE or QLIKE) against a conditionally unbiased proxy (pitfall of evaluation)
  - Many models: the model confidence set Hansen, Lunde & Nason (2011), not the lowest average loss
- VaR: coverage, independence and the DQ test Engle & Manganelli (2004)
  - with a benchmark, conditional predictive ability tests Giacomini & White (2006)
  - State the power of every backtest at your T
- ES: score the pair (VaR, ES) with FZ0
  - comparative backtests with three zones Nolde & Ziegel (2017)
  - dynamic (VaR, ES) models Patton, Ziegel & Chen (2019)
- The integrated case study (Steps 1–5) is the template

The Report and the Presentation (1/2)

The report: an 8–12-page paper

- **1. Introduction**: question, hypothesis, contribution
- **2. Literature**: 10–15 clickable references, at least 3 from the last five years
  - including Franke, Härdle & Hafner (2019), McNeil, Frey & Embrechts (2015)
- **3. Data and replication**: sources, period, cleaning rules
  - the replicated table cell by cell
- **4. Design**: plan, identification, power, benchmark, loss, tests
- **5. Results** with their uncertainty
- **6. Robustness**: specification curve with the correction
- **7. Conclusion**: the answer in one sentence, and its limits

The Report and the Presentation (2/2)

The presentation (about 12 minutes): the slides, in order

- Question (1 slide)
- Data and one chart that shows the problem (2)
- Method and benchmark (2)
- Main result with its uncertainty (2–3)
- Robustness, limits, answer (2)
- One message per slide, readable axes, legend below the chart
- Every member presents and can answer questions on the code
  - then the oral defence, without AI

Typical AI Errors in Financial Modelling (1/2)

- **Look-ahead**: a rolling statistic or a scaler that includes day t in the forecast for day t
  - Check: change the last observation; the forecast for that day must not move
  - shift by one day (`.shift(1)`)
- **Sign of VaR**: the 1% quantile of returns reported as VaR
  - or the 99% quantile of returns (a gain) used as the loss
  - Check: $\mathrm{VaR}_\alpha = -q_\alpha(r)$, with $q_\alpha(r)$ the α-quantile of returns, is positive whenever $q_\alpha(r) < 0$
  - a one-day VaR 1% of an index is a few percent
- **Annualisation**: volatility multiplied by 252 instead of $\sqrt{252}$
  - or 252 days used for crypto that trades every day
  - Check: over h periods, means scale with h, volatilities with $\sqrt{h}$ (i.i.d. returns)
  - use the real number of trading days (365 for crypto)

Typical AI Errors in Financial Modelling (2/2)

- **Random cross-validation (CV) on time series**
  - Shuffled K-fold trains on the future (pitfall of design, Chapter 13)
  - Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo López de Prado (2018)
- **Invented references**, wrong degrees of freedom
  - library defaults you did not choose
  - numbers stated without running code
  - Check: search every DOI on Crossref and find the claim in the paper, as in the AI-critique exercise of every seminar (Part C)

Key Results: Feedback, Beliefs and Forks

- **Feedback**: from stage 1 to stage 4 the IQR falls by 47.2%
  - the interdecile range falls by 68.2%
  - peer review narrows the dispersion but does not remove it
- **Beliefs**: the teams' predicted standard deviation of estimates is 71.7% below the realised one
- **Forks**: model choice and sampling frequency move the estimates most
  - A relative-change model gives a median annual trend of $-23\,000\%$ on daily data, -200% monthly and -4.56% yearly
  - Log differences add up exactly across frequencies and a linear trend is linear in the data, so neither has this frequency wedge
  - For a positive measure, Jensen's inequality biases the mean relative change upwards; the negative values in the paper need another mechanism (toy example below)

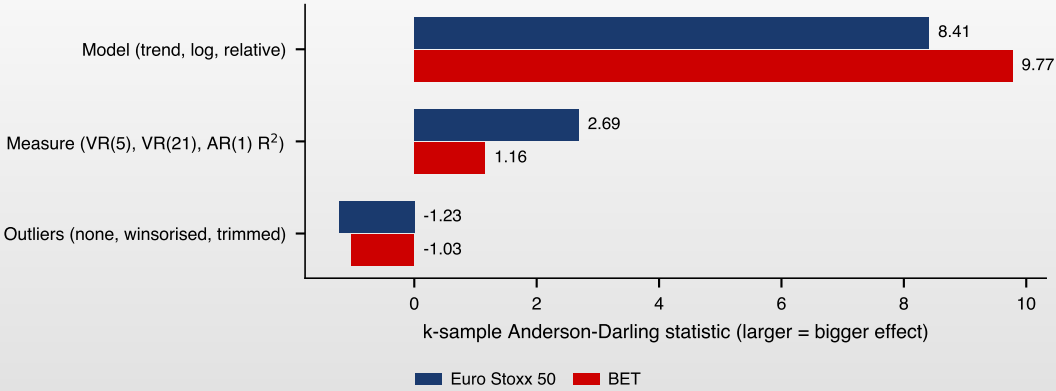
Why Relative Changes Explode: a Toy Example

- A measure that returns to its start: $m = (1,\ 0.01,\ 1)$
  - Relative changes $m_t/m_{t-1} - 1$: -99%, then $+9\,900\%$; their mean is $+4\,900.5\%$
  - Log changes: $\ln 0.01$, then $\ln 100$; their mean is exactly 0, the right answer for a measure with no trend
  - For a positive measure Jensen's inequality pushes the mean relative change up (Appendix): the course multiverse explodes upwards
- A positive measure can never fall by more than 100%
  - a trend of $-23\,000\%$ needs a measure that changes sign near zero ($m_{t-1} = 0.001$, $m_t = -0.2$ gives $-20\,100\%$)
  - or daily changes summed rather than compounded (250 changes of -1% add up to -250%)
- Lesson for your plan: inspect denominators near zero
  - pre-specify the transformation (log differences for positive measures, or a linear trend)

Which Fork Moves the Estimates? The Statistic

- k-sample Anderson–Darling statistic of the 27 estimates, grouped by the option at each fork
  - It compares the distributions of the k groups; larger means more separated
  - Standardised, so it is centred at 0 when the groups are equal: a negative value (outliers below) means less separation than chance
- A descriptive ranking, not a test
  - The 27 estimates share the same data

Which Fork Moves the Estimates?



MFM_ch19_nonstandard_errors

Which Fork Moves the Estimates? Results

- The statistic for each fork, Euro Stoxx 50 and BET
  - Model: 8.41 and 9.77
  - Measure: 2.69 and 1.16
  - Outliers: −1.23 and −1.03
- Same ranking as the paper: model choice moves the estimates most
  - on a large euro-area index and on the BET, a frontier-market index in 2002–2018
- For your project: fix the model of the yearly change in the plan (log differences or a linear trend)
  - report the whole grid with the fork that drives it

Inference by Chapter: Further Formulas

- **Chapter 3**: Fama–MacBeth standard errors ignore that the betas are estimated
  - Shanken's covariance, with $\hat{\gamma}$ the risk premia, $\hat{\Sigma}_f$ the factor covariance and $\hat{V}_{FM}$ the Fama–MacBeth (FM) covariance Shanken (1992)

$$\hat{V}_{\mathbf{sh}} = (1+c)\big(\hat{V}_{FM} - \hat{\Sigma}_f/T\big) + \hat{\Sigma}_f/T, \qquad c = \hat{\gamma}^\top \hat{\Sigma}_f^{-\mathbf{1}} \hat{\gamma}$$

  - valid for a correctly specified model; with an intercept, $\hat{\Sigma}_f$ is bordered by zeros in the row and column of the intercept
  - the empirical reference of Chapter 3: Kan–Robotti–Shanken errors, robust to misspecification, with Newey–West lags and a sensitivity to their number
- **Chapter 5**: QMLE (quasi-maximum likelihood) of GARCH Bollerslev & Wooldridge (1992)
  - A: the expected Hessian (sign changed); B: the variance of the score

$$\sqrt{T}(\hat{\theta} - \theta) \Rightarrow N\big(0, A^{-\mathbf{1}} B A^{-\mathbf{1}}\big)$$

  - in the block of the variance parameters, $A = B$ when $\kappa_z = 3$: Normal innovations, but not only
  - otherwise Hessian-only standard errors ($A^{-\mathbf{1}}$) are wrong; the factor $\sqrt{(\kappa_z - 1)/2}$ applies to the Gaussian QMLE only, not to the Student-t estimator
- **Chapter 7**: the empirical quantile, for i.i.d. returns with density $f(q_\alpha) > 0$
  - in the tail f is small, so historical VaR is noisy

$$\sqrt{T}(\hat{q}_\alpha - q_\alpha) \Rightarrow N\big(0,\ \alpha(1-\alpha)/f(q_\alpha)^{\mathbf{2}}\big)$$

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Project Ideas: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR (BET total return) autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	Does an optimised portfolio of BVB blue chips, in RON, beat 1/N out of sample?
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	How much capital does a BVB portfolio need compared with an S&P 500 portfolio?

- Each seminar gives the data, a starting method and a reference analysis discussed in class

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Project Ideas: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?
15	Does headline sentiment pay after costs?
16	Is Bitcoin an alternative asset, a risk asset or a hedge?
17	Is there an AI bubble?
18	How exposed are Romanian banks to euro-area banking stress?

- Chapter 20 also suits a forecasting project
  - realised volatility with signatures, VOLARE data

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Turning an Idea into a Project

- Replicate first: one published table or figure from a paper behind the idea (step 1)
- Narrow it: one market, one period, one number to forecast or test
  - Example: Seminar 8 becomes "Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 portfolio of the BET and EUR (EUR/RON exchange rate) in 2015–2026?"
- Extend it: a second market, a new model, a longer or more recent sample
  - Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

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Data Conventions for the Project

- Use the course data where possible
  - daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
  - Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
  - Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
  - Remove holiday rows where the vendor repeated the previous price, and weekend quotes where the market is closed
  - Annualise with the real frequency
  - Several assets: align prices on common trading days first, then compute returns
- Look for erroneous prices (bad ticks) before modelling
  - jumps of $\pm 50\%$ in adjusted prices are usually data errors

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Honest Evaluation Checklist

Before you look at the results

- Fix in the plan the hypothesis, the benchmark and the loss
  - the estimation window and the test period
  - the number of models, with their correction
  - the identification and the power

When you report

- Out-of-sample results only for claims about forecasting
- Every headline number with a standard error, CI or p-value
- Costs for trading strategies
- Subperiods and a second market for robustness
- What did **not** work, and why

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Where Projects Usually Lose Points

- A question that changes after the results are known
- In-sample fit presented as forecasting power
- No benchmark, or a weak one chosen to be beaten
- Numbers without uncertainty
- Many tests without correction
- Charts without units, legends inside the data, unexplained acronyms
- Code that does not run, or numbers in the report that the code does not produce
- AI output kept without checks
  - undeclared use, unchecked code, references that do not exist

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Nonstandard Errors: Data and Stages

- **Data**: all trades in the Euro Stoxx 50 index futures, 2002–2018
  - 720 million records
  - each side flagged as client or own account (Sec. I.A)
- **Six hypotheses**, each "no annual trend" in
  - market efficiency, realised bid–ask spread, client share of volume
  - realised spread on client orders, share of market orders among client orders
  - gross trading revenue of clients
- Each team chooses its own measure of each quantity
- **Four stages** in 2021 (Sec. I.A)
  - independent analysis (paper and code), feedback from two peer evaluators
  - the five best papers shared
  - final estimates without the code constraint
- A pre-analysis plan was filed publicly before the experiment started

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Nonstandard Errors: Quality and Inference

- **Quality**
  - Team quality: first principal component of five proxies
  - reproducibility score 0–100 by the certification agency Cascad
  - demeaned peer rating (Sec. I.B)
  - Quantile regressions of the estimates at the 10th, 25th, 50th, 75th and 90th percentiles
  - a shrinking IQR means less dispersion (Sec. I.C)
- **Inference**
  - Each estimate is tested against the median estimate, with a Bonferroni correction at a family level of 0.5% (Sec. I.C)
  - Multiverse analysis: the teams report their choice at each fork of Table V
  - The k-sample Anderson–Darling statistic ranks the forks by how much they move the estimates (Sec. II.C)

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Nonstandard Errors, Replication Details

- **Forks from Table V**
  - Outliers: none; the annual measure winsorised or trimmed at its 2.5 and 97.5 percentiles
  - Measure: variance ratio day to week, variance ratio day to month, R^2 of an autoregressive AR(1) model
  - Model: linear trend, log difference, relative difference
  - Daily data allow the day-to-week and day-to-month comparisons ($VR(5)$, $VR(21)$); the intraday measure frequencies of Table V need intraday data
- **Why relative differences explode**: $m_t = |VR - 1|$ or R^2 is close to zero in some years (paper: Fig. 7)
  - No trend, stationary m_t: the linear slope and the mean log change are centred at zero
  - but for independent m_t, $\mathbb{E}(m_t/m_{t-1}) = \mathbb{E}m \cdot \mathbb{E}(1/m) \geq 1$ (Jensen), so the mean relative change is positive
- **Anderson–Darling statistic**: the 27 estimates are strongly correlated
  - so tabulated critical values and p-values do not apply

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The Discovery Loop, Prompts (1/2)

- **Week 1, the question**: sharpen a project idea into H_0/H_1 with a replication target
  - "Here is our project idea from Part C of Seminar N. Give three sharper, testable versions, the published table each would replicate and the series from the course data it needs."
- **Weeks 1–2, literature and target**: the number to replicate is one you have seen in the PDF
  - "List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample."
  - Resolve every DOI on Crossref
  - Search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves
- **Week 2, hypotheses and pre-registration**, recorded in a dated commit before the first result
  - "For our H0 and H1, list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction."

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The Discovery Loop, Prompts (2/2)

- **Weeks 2–3, code and replication**: the replicated table becomes a test in the repository
  - "Write a function that reproduces every cell of the published table from the course data and an assertion with the tolerance we accept."
- **Weeks 4–5, extension and robustness**: run the whole pre-registered grid
  - "Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?"
- **Before the defence, critique**: the assistant asks, the team answers without AI, as at the defence itself
  - "Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them."
- **Report**: the distribution of results over the grid, not the best cell
  - prompts in `AI_USE.md`
  - caught errors in `AI_ERRORS.md`

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What the Researcher Must Check

- **The replication target exists**
  - number, table and sample taken from the paper, not from the assistant's summary
  - the report cites the page and the table, and a second member finds the same number there
- **The plan came first**
  - in the commit history, the pre-registration of the extension precedes the first extension results file
- **Forking paths** (more specifications tried than reported)
  - every run writes its configuration and result to a log
  - the report states their number and corrects the family
- **Reproducibility** (no numbers entered by hand)
  - a fresh clone and one command regenerate every number to the reported digit
- **Understanding**: each member can derive the key formula and defend every choice
  - the test is an internal mock defence, without AI, one week before the real one

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Acronyms used in this material (1/3)

- **ACF**: AutoCorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ASE**: Academia de Studii Economice din București — *the Bucharest University of Economic Studies*
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BSADF**: Backward Sup Augmented Dickey–Fuller (statistic)
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAGR**: Compound Annual Growth Rate
- **CAPM**: Capital Asset Pricing Model
- **CI**: Confidence Interval
- **CoVaR**: Conditional Value-at-Risk (of the system, given an institution)
- **COVID-19**: COronaVIrus Disease 2019
- **CV**: Cross-Validation
- **DCC**: Dynamic Conditional Correlation
- **DeFi**: Decentralised Finance
- **DM**: Diebold–Mariano (test)
- **DOI**: Digital Object Identifier
- **DQ**: Dynamic Quantile (test)
- **EDGE**: Efficient Discrete Generalized Estimator (of the bid-ask spread)
- **EEM**: iShares MSCI Emerging Markets ETF
- **EFA**: iShares MSCI EAFE ETF (developed markets outside the US and Canada)

Acronyms used in this material (2/3)

- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **EUR**: euro, international currency code
- **EVT**: Extreme Value Theory
- **FDR**: False Discovery Rate
- **FHS**: Filtered Historical Simulation
- **FM**: Fama–MacBeth (regressions)
- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-EVT**: GARCH filter combined with Extreme Value Theory for the tail
- **GBM**: Geometric Brownian Motion
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GLD**: SPDR Gold Shares
- **GMV**: Global Minimum Variance (portfolio)
- **GRS**: Gibbons–Ross–Shanken (test)
- **GSADF**: Generalised Sup Augmented Dickey–Fuller (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HS**: Historical Simulation
- **HYG**: iShares iBoxx High Yield Corporate Bond ETF
- **IEF**: iShares 7–10 Year Treasury Bond ETF
- **IGARCH**: Integrated GARCH
- **IQR**: InterQuartile Range

Acronyms used in this material (3/3)

- **JB**: Jarque–Bera (test)
- **LLM**: Large Language Model
- **LQD**: iShares iBoxx Investment Grade Corporate Bond ETF
- **LR**: Likelihood Ratio
- **LRMES**: Long-Run Marginal Expected Shortfall
- **LSTM**: Long Short-Term Memory (network)
- **MCS**: Model Confidence Set
- **MDE**: Minimum Detectable Effect
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **PDF**: Portable Document Format
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **RON**: Romanian leu, international currency code
- **RV**: Realised Variance
- **SADF**: Sup Augmented Dickey–Fuller (test)
- **SE**: Standard Error
- **SML**: Security Market Line
- **SPY**: SPDR S&P 500 ETF Trust
- **SR**: Sharpe Ratio
- **SRISK**: Systemic RISK (capital shortfall measure)
- **TLT**: iShares 20+ Year Treasury Bond ETF
- **US**: United States
- **USD**: US dollar, international currency code
- **UTC**: Coordinated Universal Time
- **VaR**: Value-at-Risk
- **VIX**: CBOE Volatility IndeX
- **VOLARE**: VOLatility Archive for Realized Estimates (database)
- **VR**: Variance Ratio
- **VRP**: Variance Risk Premium

References (1/6)

- Abadie, A., Diamond, A., Hainmueller, J. (2010). Synthetic control methods for comparative case studies: estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490), 493–505.
- Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- Andrews, D.W.K. (1999). Estimation when a parameter is on a boundary. *Econometrica*, 67(6), 1341–1383.
- Athreya, K.B. (1987). Bootstrap of the mean in the infinite variance case. *Annals of Statistics*, 15(2), 724–731.
- Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Barndorff-Nielsen, O.E., Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300.
- Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

References (2/6)

- Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.
- Brownlees, C., Engle, R.F. (2017). SRISK: a conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1), 48–79.
- Campbell, J.Y., Serfaty-de Medeiros, K., Viceira, L.M. (2010). Global currency hedging. *Journal of Finance*, 65(1), 87–121.
- Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.
- Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Engle, R. (2002). Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

References (3/6)

- Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Fissler, T., Ziegel, J.F. (2016). Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). Applied Quantitative Finance, 3rd ed. Springer.
- Harvey, C.R. (2017). Presidential address: the scientific outlook in financial economics. *Journal of Finance*, 72(4), 1399–1440.

References (4/6)

- Harvey, C.R., Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5), 2503–2553.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- Ledoit, O., Wolf, M. (2004). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lo, A.W., MacKinlay, A.C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.

References (5/6)

- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- McNeil, A.J., Frey, R., Embrechts, P. (2015). Quantitative Risk Management: Concepts, Techniques and Tools, revised ed. Princeton University Press.
- Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). Nonstandard errors. *Journal of Finance*, 79(3), 2339–2390.
- Nelson, D.B. (1990). Stationarity and persistence in the GARCH(1,1) model. *Econometric Theory*, 6(3), 318–334.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Nolde, N., Ziegel, J.F. (2017). Elicitability and backtesting: perspectives for banking regulation. *Annals of Applied Statistics*, 11(4), 1833–1874.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

References (6/6)

- Patton, A.J., Ziegel, J.F., Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413.
- Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- Phillips, P.C.B., Shi, S., Yu, J. (2015). Testing for multiple bubbles: historical episodes of exuberance and collapse in the S&P 500. *International Economic Review*, 56(4), 1043–1078.
- Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). Specification curve analysis. *Nature Human Behaviour*, 4(11), 1208–1214.
- Stambaugh, R.F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375–421.
- Tsay, R.S. (2010). Analysis of Financial Time Series, 3rd ed. Wiley.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.
- Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). A tale of two time scales: determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411.